

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\106-D.pvdb

Date of Analysis : May 11, 2026 10:50am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 5542.000 mm OD	
Vessel Design Length, Tangent to Tangent	1850.00 mm
Distance of Bottom Tangent above Grade	5500.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	5200.00 mm
Internal Design Temperature	454 °C
Internal Design Pressure	4621.000 KPa.
External Design Temperature	454 °C
External Design Pressure	103.419 KPa.
Maximum Allowable Working Pressure	4659 KPa.
External Max. Allowable Working Pressure	911 KPa.
Shop Test Pressure	6705.829 KPa.
Field Test Pressure	6475.703 KPa.
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-16.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	+IT
Skirt	SA-516 70	...	...	K02700	
Skirt	SA-387 11	2	...	K11789	+IT
Nozzle	SA-182 F11	2	...	K11572	+IT
Nozzle Flg	SA-182 F11	2	...	K11572	
Basering	SA-36	...	...	K02600	
Base Bolting	SA-307 A	...	6 ≤ t ≤ 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-3	...	...	...	1.6000	N/A	No
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
Bottom head	4621.000	103.42	4659.000	3.0000	N/A	No
Shell	4621.000	103.42	4670.000	3.0000	N/A	No
Top head	4621.000	103.42	4659.000	3.0000	N/A	No

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-4000.0	1200.0	24.0	24.0	...	...	0.70	0.70
Skirt	-1000.0	3000.0	24.0	24.0	...	...	0.70	0.70
Skirt	0.0	1000.0	24.0	24.0	...	...	0.70	0.50



Sphere	0.0	0.0	54.0	49.0	48.6	12.1	1.00	1.00
Cylinder	1850.0	1850.0	96.0	96.0	95.0	19.7	1.00	1.00
Sphere	1850.0	0.0	54.0	49.0	48.6	12.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2_600	WRC-107/537	0.989	n/a	Passed
N1_750	WRC-107/537	0.469	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	600	3641.02	48761.5	19905.7	307441
20	30	2700	3339.59	48635.2	15717.2	249002
30	40	4700	2391.42	47812.9	7120.55	104327
40	50	5199.98	2140.23	47403.3	4854.74	56719.7
50	60	6125	2140.22	32642.5	4854.68	56718.5
60	70	7049.98	1334.09	12308.1	1640.92	15138.8

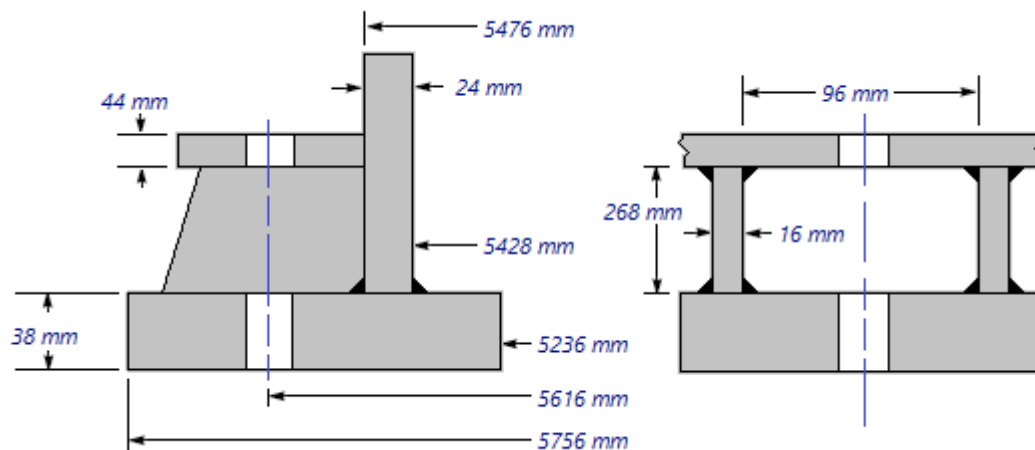
Abs Max of the all of the Stress Ratio's : 0.9783

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	38.0000	mm
Inside Diameter of Basing	5236.0000	mm
Outside Diameter of Basing	5756.0000	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	5616.0000	mm
Number of Bolts	24	

Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	130.0000	mm
Height of Gussets	268.0000	mm
Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	140.0000	mm

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:



Total Wind Shear on Support	11373. Kgf
Total Earthquake Shear on Support	56493. Kgf
Wind Moment on Support	121950. Kg-m.
Earthquake Moment on Support	409486. Kg-m.

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	111197.9 kg.
Shop Test - Fabricated + water (full)	235088.9 kg.
Shipping - Fab. + rem. intls.+ shipping app.	203978.8 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	204978.8 kg.
Empty - Fab. + intls. + details + wghts.	204978.8 kg.
Operating - Empty + operating liquid (no ca)	310978.8 kg.
Field Test - Empty weight + test medium (full)	254547.5 kg.

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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2_600	4659	OK	...	OK	13.84	OK	Passed
U1_600	4659	OK	...	OK	13.84	OK	Passed
K5_75	4659	OK	...	OK	8.02	OK	Passed
K4_50	4670	...	...	OK	7.80	OK	No Calc[*]
M1_600	4659	OK	...	...	...	OK	Passed
K3_50	4659	OK	...	OK	7.52	OK	Passed
K3_50	4659	...	...	OK	7.52	OK	No Calc[*]
K1_50	4659	OK	...	OK	7.52	OK	Passed
K1_50	4659	...	...	OK	7.52	OK	No Calc[*]
N1_750	4659	OK	...	OK	17.18	OK	Passed
K2_50	4659	OK	...	OK	7.52	OK	Passed
K2_50	4659	...	...	OK	7.52	OK	No Calc[*]
K6_75	4659	OK	...	OK	8.02	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4659 Nozzle : K675
 Minimum MAWP Shells/Flanges : 4659 Element : Top head
 Minimum MAPnc Shells/Flanges : 5199 Element : Shell

Computed Vessel M.A.W.P. : 4659 KPa.

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
40	N2_600	0.000	0.000	1132.000
40	U1_600	0.000	180.000	1271.053
40	K5_75	0.000	0.000	187.000
50	K4_50	900.000	0.000	270.400
60	M1_600	0.000	0.000	1148.000
60	K3_50	0.000	0.000	162.000
60	K1_50	0.000	240.000	162.000
60	N1_750	0.000	0.000	1512.000
60	K2_50	0.000	120.000	162.800
60	K6_75	0.000	180.000	188.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K3_50	50 mm	Actual	WNF	70.0	10.000	...	...	2332	600
K1_50	50 mm	Actual	WNF	70.0	10.000	...	...	6058	600
K4_50	50 mm	Actual	LWN	84.4	16.800	...	...	446	600
K2_50	2 in	LWN	WNF	70.8	10.000	...	...	5358	600
K5_75	75 mm	Actual	WNF	95.0	10.000	...	...	619	600
K6_75	76 mm	Actual	WNF	96.2	10.000	...	...	2174	600
N2_600	600 mm	Actual	WNF	600.0	20.000	...	...	648	600
U1_600	600 mm	Actual	WNF	600.0	20.000	...	...	2549	600
M1_600	600 mm	Actual	WNF	600.0	16.000	...	...	1291	600
N1_750	750 mm	Actual	WNF	790.0	20.000	...	...	1129	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K3_50	SA-182 F11	49.000	10.000	...	...	...
K1_50	SA-182 F11	49.000	10.000	...	...	...
K4_50	SA-182 F11	96.000	10.000	...	...	...
K2_50	SA-182 F11	49.000	10.000	...	...	14.000
K5_75	SA-182 F11	49.000	10.000	...	...	...
K6_75	SA-182 F11	49.000	10.000	...	...	...
N2_600	SA-182 F11	49.000	10.000	...	...	...
U1_600	SA-182 F11	49.000	10.000	...	...	...
M1_600	SA-182 F11	49.000	10.000	...	...	...
N1_750	SA-182 F11	49.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K3_50	...	0.0	2000.00	0.00	Top head
K1_50	...	240.0	1000.00	5000.00	Top head
K4_50	900.000	0.0	350.00	0.00	Shell
K2_50	...	120.0	300.00	5000.00	Top head
K5_75	...	0.0	500.00	0.00	Bottom head
K6_75	...	180.0	2000.00	0.00	Top head
N2_600	...	0.0	300.00	300.00	Bottom head
U1_600	...	180.0	1500.00	1000.00	Bottom head
M1_600	...	0.0	1000.00	0.00	Top head
N1_750	...	0.0	1000.00	0.00	Top head

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 54.0mm THK X 5398.0mm ID X 0.0mm	SA-387 11
2	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 2500.0mm	...
1	INSULATION: 2675mm X 150mm THK	
1	LADDER: 3000.0mm	...
1	PACKING: 1850.000mm	...
1	INSULATION: 1850mm X 150mm THK	
1	PLATFORM: 1000mm X 5000mm WIDE	
1	LADDER: 5000.0mm	...
1	INSULATION: 2693mm X 150mm THK	
2	CLASS 600 GR 1.9, 600.0mm BLIND FLANGE(S)	SA-182 F11
2	CLASS 600 GR 1.9, 50.8mm BLIND FLANGE(S)	SA-182 F11
1	CLASS 600 GR 1.9, 50.0mm BLIND FLANGE(S)	SA-182 F11
1	BASERING: 5756.0mm OD X 5236.0 ID X 38.0mm	SA-36
24	BASERING BOLTS: 48.0mm DIA	SA-307 A
24	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 5756.0mm OD X 5476.0 ID X 44.0mm	...
48	BASERING GUSSET PLATES: 130.0mm X 130.0mm X 268.0mm X 16.0mm	...
4	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 3078.9mm X 1200.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 3126.3mm X 1200.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 1610.5mm X 1200.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 3789.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 3836.8mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 1847.3mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 3789.4mm X 1000.0mm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 3836.8mm X 1000.0mm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 1847.3mm X 1000.0mm	SA-387 11
2	CYLINDRICAL SEGMENT 96.0mm THK X 6676.2mm X 1850.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 96.0mm THK X 3454.8mm X 1850.0mm	SA-387 11
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom head	[9]	!	-12	-16		0.927	49.000	1.00	Yes
Shell	[8]	!	-12	-16		0.926	96.000	1.00	Yes
Top head	[9]	!	-12	-16		0.927	49.000	1.00	Yes
N2600	[1]	!	-12	-16		0.920	49.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
U1600	[1]	!	-12	-16		0.920	49.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K575	[1]	!	-12	-16		0.920	49.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K450	[1]	!	-12	-104		0.070	16.800	1.00	No
Nozzle Flg	[5]	B	-18	-48					
M1600	[1]	!	-12	-16		0.920	49.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K350	[1]	!	-12	-16		0.920	49.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K150	[1]	!	-12	-16		0.920	49.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
N1750	[1]	!	-7	-21		0.760	20.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K250	[1]	!	-12	-16		0.920	49.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K675	[1]	!	-12	-16		0.920	49.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Warmest MDMT:			-7	-16					

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -16.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.





Class From To : Basic Element Checks.

Note 30 50 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	4621	KPa.
Design Internal Temperature	454.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	Km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		300	mm
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm
Distance Upwind of Crest	Lh	0	mm
Distance from Crest to the Vessel	x	0	mm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm

Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		2.000	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		2.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-3	
Distance "FROM" to "TO"	1200	mm
Skirt Outside Diameter	5476	mm
Diameter of Skirt at Base	5476	mm
Skirt Thickness	24	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	50	°C
Design Temperature External Pressure	50	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	255.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing lug	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1500	Kgf
Offset from Element Centerline	0	mm

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	5476	mm
Diameter of Skirt at Base	5476	mm
Skirt Thickness	24	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	50	°C
Design Temperature External Pressure	50	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Platform	
Detail ID	PL-3	
Dist. from "FROM" Node / Offset dist	1200	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	25160	cm ²
Platform Weight	1575.9	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm



Platform Force Coefficient	1.0	
Ladder Layout Angle	90.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	2500	mm
Unit Weight of Ladder	0.03773	Kgf/mm
Platform Length (top head platform)	0	mm

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	5476	mm
Diameter of Skirt at Base	5476	mm
Skirt Thickness	24	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	454	°C
Design Temperature External Pressure	454	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-387 11 +IT	
Allowable Stress, Ambient	148	N./mm ²
Allowable Stress, Operating	137.29	N./mm ²
Allowable Stress, Hydrotest	278.99	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	16.002	mm
Yield Stress, Operating	218.04	N./mm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Spherical	
Description	Bottom head	
Distance "FROM" to "TO"	0	mm
Inside Diameter	5398	mm
Element Thickness	49	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	54	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4621	KPa.
Design Temperature Internal Pressure	454	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	454	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-2675	mm
Height/Length of Insulation	2675	mm
Thickness of Insulation	150	mm
Density	0.000352	kg./cm ³



Element From Node 40
 Detail Type Nozzle
 Detail ID N2600
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 600 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kg
 Grade of Attached Flange GR 1.9
 Nozzle Matl SA-182 F11 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID U1600
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 600 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 180.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 0 Kg
 Grade of Attached Flange GR 1.9
 Nozzle Matl SA-182 F11 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K575
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 75 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kg
 Grade of Attached Flange GR 1.9
 Nozzle Matl SA-182 F11 +IT

Element From Node 40
 Detail Type Weight
 Detail ID Closure plate
 Dist. from "FROM" Node / Offset dist -2693 mm
 Miscellaneous Weight 250 Kg
 Offset from Element Centerline 0 mm

Element From Node 40
 Detail Type Weight
 Detail ID Inert Balls
 Dist. from "FROM" Node / Offset dist 0 mm
 Miscellaneous Weight 77000 Kg
 Offset from Element Centerline 0 mm

Element# 5/6
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description Shell
 Distance "FROM" to "TO" 1850 mm
 Element Outside Diameter 5542 mm
 Element Thickness 96 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 96 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4621 KPa.
 Design Temperature Internal Pressure 454 °C
 Design External Pressure 103.42 KPa.



Design Temperature External Pressure	454	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Platform	
Detail ID	PL-1	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	42192	cm ²
Platform Weight	6054	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	3000	mm
Unit Weight of Ladder	0.03773	Kgf/mm
Platform Length (top head platform)	0	mm
Element From Node	50	
Detail Type	Packing	
Detail ID	Catalyst	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	1850	mm
Density	0.001761	kg./cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1850	mm
Thickness of Insulation	150	mm
Density	0.000352	kg./cm ³
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K450	
Dist. from "FROM" Node / Offset dist	900	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	+IT
Element From Node	50	
Detail Type	Weight	
Detail ID	SF wt2	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	2000	Kgf
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Clip wt	
Dist. from "FROM" Node / Offset dist	899.97	mm
Miscellaneous Weight	2670	Kgf
Offset from Element Centerline	0	mm



Element From Node	50
Detail Type	Weight
Detail ID	Trunnion 1
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	1000 Kgf
Offset from Element Centerline	3000 mm

Element From Node	50
Detail Type	Weight
Detail ID	Trunnion-2
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	1000 Kgf
Offset from Element Centerline	3000 mm

Element From Node	50
Detail Type	Weight
Detail ID	Misc 10%
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	10000 Kgf
Offset from Element Centerline	925 mm

Element#	6/6
Element From Node	60
Element To Node	70
Element Type	Spherical
Description	Top head
Distance "FROM" to "TO"	0 mm
Inside Diameter	5398 mm
Element Thickness	49 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	54 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4621 KPa.
Design Temperature Internal Pressure	454 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	454 °C
Effective Diameter Multiplier	1.22
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Platform
Detail ID	PL-3
Dist. from "FROM" Node / Offset dist	3000 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	70320 cm ²
Platform Weight	4440.8 Kgf
Platform Railing Weight	0 Kgf/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	5000 mm
Platform Height	1000 mm
Platform Clearance or End Offset	2771 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	90.0
Ladder Start Elevation	0 mm
Ladder End Elevation	5000 mm
Unit Weight of Ladder	0.03773 Kgf/mm
Platform Length (top head platform)	5000 mm

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2693 mm



Thickness of Insulation 150 mm
Density 0.000352 kg./cm³

Element From Node 60
Detail Type Nozzle
Detail ID M1600
Dist. from "FROM" Node / Offset dist 1500 mm
Nozzle Diameter 600 mm
Nozzle Schedule None
Nozzle Class 600
Layout Angle 0.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.9
Nozzle Matl SA-182 F11 +IT

Element From Node 60
Detail Type Nozzle
Detail ID K350
Dist. from "FROM" Node / Offset dist 2500 mm
Nozzle Diameter 50 mm
Nozzle Schedule None
Nozzle Class 600
Layout Angle 0.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.9
Nozzle Matl SA-182 F11 +IT

Element From Node 60
Detail Type Nozzle
Detail ID K150
Dist. from "FROM" Node / Offset dist 1500 mm
Nozzle Diameter 50 mm
Nozzle Schedule None
Nozzle Class 600
Layout Angle 240.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.9
Nozzle Matl SA-182 F11 +IT

Element From Node 60
Detail Type Nozzle
Detail ID N1750
Dist. from "FROM" Node / Offset dist 0 mm
Nozzle Diameter 750 mm
Nozzle Schedule None
Nozzle Class 600
Layout Angle 0.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.9
Nozzle Matl SA-182 F11 +IT

Element From Node 60
Detail Type Nozzle
Detail ID K250
Dist. from "FROM" Node / Offset dist 1500 mm
Nozzle Diameter 50.8 mm
Nozzle Schedule None
Nozzle Class 600
Layout Angle 120.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.9
Nozzle Matl SA-182 F11 +IT

Element From Node 60
Detail Type Nozzle



Detail ID	K675	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	76.2	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11 +IT	
Element From Node	60	
Detail Type	Weight	
Detail ID	N1vane distrib	
Dist. from "FROM" Node / Offset dist	2675	mm
Miscellaneous Weight	300	Kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	M1Davit	
Dist. from "FROM" Node / Offset dist	2400	mm
Miscellaneous Weight	100	Kgf
Offset from Element Centerline	1500	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	SF wt-1	
Dist. from "FROM" Node / Offset dist	-0.02601	mm
Miscellaneous Weight	2000	Kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	K2Internals	
Dist. from "FROM" Node / Offset dist	2400	mm
Miscellaneous Weight	350	Kgf
Offset from Element Centerline	1500	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	K1Internals	
Dist. from "FROM" Node / Offset dist	2400	mm
Miscellaneous Weight	350	Kgf
Offset from Element Centerline	1500	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	Inert Balls	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	29000	Kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	N1 insert	
Dist. from "FROM" Node / Offset dist	2675	mm
Miscellaneous Weight	850	Kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	For./Mom.	
Detail ID	N2F/M	
Dist. from "FROM" Node / Offset dist	2693	mm
Force in X Direction	5467	Kgf
Force in Y Direction	-7568	Kgf
Force in Z Direction	5467	Kgf
Moment about X Axis	12615	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	12615	Kg-m.

PV Elite 28 Licensee: SL User
FileName : 106-D
Input Echo:

Step: 1 10:50am May 11,2026

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HEXAGON

Force/Moment Combination Method

SRSS

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**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-3	...	...	1200	...	1200
Skirt-2	...	...	4200	...	3000
Skirt-1	...	...	5200	...	1000
Bottom head	...	...	5200	...	...
Shell	...	...	7050	...	1850
Top head	...	...	7050	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Skirt-3	...	...	24	1.6	5476	...	1
Skirt-2	...	...	24	1.6	5476	...	2
Skirt-1	...	...	24	1.6	5476	...	3
Bottom head		4621	54	3	5398	137.29	4
Shell		4621	96	3	5542	137.29	5
Top head		4621	54	3	5398	137.29	6

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-3	...	...	No Calc	No Calc	24	No Calc	1
Skirt-2	...	...	No Calc	No Calc	24	No Calc	2
Skirt-1	...	...	No Calc	No Calc	24	No Calc	3
Bottom head		4621	4658.5	5353.98	49	48.6284	4
Shell		4621	4670.21	5199.04	96	95.0331	5
Top head		4621	4658.5	5353.98	49	48.6284	6

Minimum

4659

5199

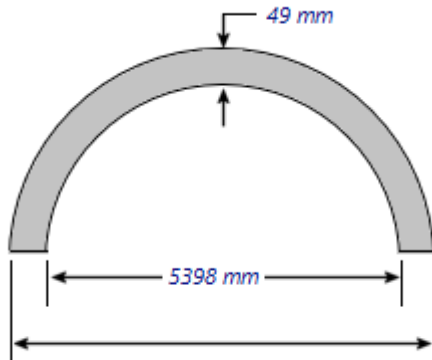
MAWP: 4659 KPa., limited by: Top head.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom head

Spherical Head Design Information:					
Design Pressure	P	4621.000 KPa.	Int. Design Temperature	454 °C	
Material		SA-387 11	Ext. Design Temperature	454 °C	
Length		0.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	137.3 N./mm ²	
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²	
Inside Diameter	Di	5398.000 mm	Actual Stress	136.2 N./mm ²	
Spec. Min. Thk t,e, or sa		49.0000 mm	Spec. Nominal Thk.	54.000 mm	
Surface Area		476203.2 cm ²	SG of Contents	0.000	
Element Volume		41185568.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		19542.0 Kgf	Operating Weight	19542.0 Kgf	
			Comp. All. Ope. Stress	65.3 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 67.475 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 33.737 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= (4621.0 \cdot 2702.0) / (2 \cdot 137.29 \cdot 1.0 - 0.2 \cdot 4621.0)$
 $= 45.6284 + 3.0000 = 48.6284 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 137.29 \cdot 1.0 \cdot 46.0) / (2702.0 + 0.2 \cdot 46.0)$
 $= 4659 \text{ KPa.}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 148.0 \cdot 1.0 \cdot 49.0) / (2699.0 + 0.2 \cdot 49.0)$
 $= 5354 \text{ KPa.}$

Actual stress at given pressure and thickness, corroded [Sact]:



$$\begin{aligned} &= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (4621.0 \cdot (2702.0 + 0.2 \cdot 46.0)) / (2 \cdot 1.0 \cdot 46.0) \\ &= 136.187 \text{ N./mm}^2 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.486 %

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 67.475 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 33.737 mm

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Bottom head

Spherical Head From 40 to 50 Ext. Chart: CS-2 at 454 °C

Elastic Modulus from Chart: CS-2 at 454 °C: 150421952 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
46.000	5496.00	119.48	0.0020924	54.42

$$MAEP = B / ((Do/t)/2) = 54.4228 / (119.4782/2) = 910.9543 \text{ KPa.}$$

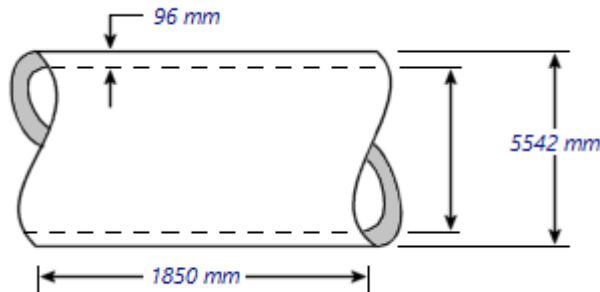
Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
9.115	5496.00	602.99	0.0004146	31.18

$$MAEP = B / ((Do/t)/2) = 31.1843 / (602.9911/2) = 103.4259 \text{ KPa.}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	4621.000 KPa.	Int. Design Temperature	454 °C	
Material		SA-387 11	Ext. Design Temperature	454 °C	
Length		1850.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.3 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	148.0 N./mm ²
Outside Diameter	Do	5542.000 mm	Actual Stress	135.8 N./mm ²	
Spec. Min. Thk t _e , or s _a		96.0000 mm	Spec. Nominal Thk.	96.000 mm	
Surface Area		322098.1 cm ²	SG of Contents	0.000	
Element Volume		41595564.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		23550.5 Kgf	Operating Weight	23550.5 Kgf	
			Comp. All. Ope. Stress	74.7 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (4621.0 \cdot 2771.0) / (137.29 \cdot 1.0 + 0.4 \cdot 4621.0)$$

$$= 92.0331 + 3.0000 = 95.0331 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (137.29 \cdot 1.0 \cdot 93.0) / (2771.0 - 0.4 \cdot 93.0)$$

$$= 4670 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 96.0) / (2771.0 - 0.4 \cdot 96.0)$$

$$= 5199 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (4621.0 \cdot (2771.0 - 0.4 \cdot 93.0)) / (1.0 \cdot 93.0)$$

$$= 135.845 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 1.763 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 454 °C

Elastic Modulus from Chart: CS-2 at 454 °C: 150421952 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
93.000	5542.00	3649.33	59.59	0.6585	0.0046564	63.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*63.3486)/(3*59.5914) = 1417.3177 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.685	5542.00	3649.33	332.15	0.6585	0.0003426	25.77

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7652)/(3*332.1483) = 103.4224 \text{ KPa.}$$

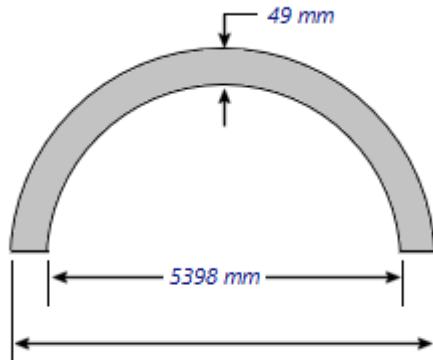
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
93.000	5542.00	102892.00	59.59	18.5659	0.0003138	23.60

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.6004)/(3*59.5914) = 528.0193 \text{ KPa.}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top head

Spherical Head Design Information:					
Design Pressure	P	4621.000 KPa.	Int. Design Temperature	454 °C	
Material		SA-387 11	Ext. Design Temperature	454 °C	
Length		0.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	137.3 N./mm ²	
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²	
Inside Diameter	Di	5398.000 mm	Actual Stress	136.2 N./mm ²	
Spec. Min. Thk t,e, or sa		49.0000 mm	Spec. Nominal Thk.	54.000 mm	
Surface Area		476203.2 cm ²	SG of Contents	0.000	
Element Volume		41185568.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		19542.0 Kgf	Operating Weight	19542.0 Kgf	
			Comp. All. Ope. Stress	65.3 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 67.475 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 33.737 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= (4621.0 \cdot 2702.0) / (2 \cdot 137.29 \cdot 1.0 - 0.2 \cdot 4621.0)$
 $= 45.6284 + 3.0000 = 48.6284 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 137.29 \cdot 1.0 \cdot 46.0) / (2702.0 + 0.2 \cdot 46.0)$
 $= 4659 \text{ KPa.}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 148.0 \cdot 1.0 \cdot 49.0) / (2699.0 + 0.2 \cdot 49.0)$
 $= 5354 \text{ KPa.}$

Actual stress at given pressure and thickness, corroded [Sact]:



$$\begin{aligned} &= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (4621.0 \cdot (2702.0 + 0.2 \cdot 46.0)) / (2 \cdot 1.0 \cdot 46.0) \\ &= 136.187 \text{ N./mm}^2 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.486 %

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 67.475 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 33.737 mm

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Top head

Spherical Head From 60 to 70 Ext. Chart: CS-2 at 454 °C

Elastic Modulus from Chart: CS-2 at 454 °C: 150421952 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
46.000	5496.00	119.48	0.0020924	54.42

$$MAEP = B / ((Do/t)/2) = 54.4228 / (119.4782/2) = 910.9543 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
9.115	5496.00	602.99	0.0004146	31.18

$$MAEP = B / ((Do/t)/2) = 31.1843 / (602.9911/2) = 103.4259 \text{ KPa.}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6528.258 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6475.703 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6705.829 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5523.911 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6608.030 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6608.030 KPa.
App 27-4	= M.A.W.P.	4658.502 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	186.8	279.0	0.670	6758.74
Shell	192.4	279.0	0.690	6760.15
Top head	186.8	279.0	0.670	6758.74

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2_600	Nozzle	138.00	126.04	1.095
U1_600	Nozzle	138.00	126.04	1.095
K5_75	Nozzle	138.00	126.04	1.095
K4_50	Nozzle	138.00	126.00	1.095
M1_600	Nozzle	138.00	126.04	1.095
K3_50	Nozzle	138.00	126.04	1.095
K1_50	Nozzle	138.00	126.04	1.095
N1_750	Nozzle	138.00	126.00	1.095
K2_50	Nozzle	138.00	126.00	1.095
K6_75	Nozzle	138.00	126.00	1.095

Minimum 1.095

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom head	148.00	137.29	1.078
Shell	148.00	137.29	1.078
Top head	148.00	137.29	1.078

Minimum 1.078

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2_600	98.68	247.50	0.399
U1_600	98.68	247.50	0.399
K5_75	29.40	247.50	0.119
K4_50	14.28	247.50	0.058
M1_600	124.03	247.50	0.501
K3_50	20.95	247.50	0.085
K1_50	20.95	247.50	0.085
N1_750	130.79	247.50	0.528
K2_50	21.22	247.50	0.086
K6_75	29.81	247.50	0.120

Elements Suitable for Test Pressure.





Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6528.258 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6475.703 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6714.148 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5523.911 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6608.030 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6608.030 KPa.
App 27-4	= M.A.W.P.	4658.502 KPa.

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 4621.0 • 1.078
 = 6475.703 KPa.

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	192.9	279.0	0.692	6546.75
Shell	191.7	279.0	0.687	6520.29
Top head	191.6	279.0	0.687	6502.16

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2_600	Nozzle	138.00	126.04	1.095
U1_600	Nozzle	138.00	126.04	1.095
K5_75	Nozzle	138.00	126.04	1.095
K4_50	Nozzle	138.00	126.00	1.095
M1_600	Nozzle	138.00	126.04	1.095
K3_50	Nozzle	138.00	126.04	1.095
K1_50	Nozzle	138.00	126.04	1.095
N1_750	Nozzle	138.00	126.00	1.095
K2_50	Nozzle	138.00	126.00	1.095
K6_75	Nozzle	138.00	126.00	1.095

Minimum 1.095

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom head	148.00	137.29	1.078
Shell	148.00	137.29	1.078
Top head	148.00	137.29	1.078

Minimum 1.078

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2_600	95.59	247.50	0.386
U1_600	95.59	247.50	0.386
K5_75	28.48	247.50	0.115
K4_50	13.77	247.50	0.056
M1_600	119.32	247.50	0.482
K3_50	20.16	247.50	0.081
K1_50	20.16	247.50	0.081
N1_750	125.82	247.50	0.508



K2_50	20.42	247.50	0.082
K6_75	28.68	247.50	0.116

Elements Suitable for Test Pressure.

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**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 1, 2025****External Pressure Calculations:**

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	5200	...	...	No Calc	No Calc
20	30	5200	...	...	No Calc	No Calc
30	40	5200	...	...	No Calc	No Calc
40	50	No Calc	5496	46	0.0020924	54.4228
50	60	3649.33	5542	93	0.0046564	63.3486
60	70	No Calc	5496	46	0.0020924	54.4228

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	49	12.1146	103.419	910.954
50	60	96	19.6853	103.419	1417.32
60	70	49	12.1146	103.419	910.954

Minimum

911

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	5200	No Calc	No Calc	No Calc
20	30	5200	No Calc	No Calc	No Calc
30	40	5200	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	3649.33	102892	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	6195.11	...	5940.23	...	...
20	30	9558.05	...	8920.85	...	...
30	40	3186.01	...	2973.64	...	...
40	50	19542	41185568	18476.6	41323056	...
50	60	23550.5	41595564	22827.1	41688916	...
60	70	19542	41185568	18476.6	41323056	...
Total		81573	0.124E+09	77615	0.124E+09	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	1500	...	...	...	Tailing lug
20	Plat	1575.94	2355.74	1200	-2355.74	PL-3
40	Insl	2819.48	...	-1337.5	...	Ins: 10
40	Nozl	1020.66	...	-2674.5	...	N2_600
40	Nozl	2998.83	-1500	-2469.29	...	U1_600
40	Nozl	142.078	1500	-4878.6	...	K5_75
40	Wght	250	...	-2693	...	Closure plate
40	Wght	77000	...	...	...	Inert Balls
50	Plat	6054.05	0.32426E-04	...	...	PL-1
50	Pack	73322.4	...	925	...	Catalyst
50	Insl	1748.75	...	925	...	Ins: 20
50	Nozl	21.5506	2850.01	900	...	K4_50
50	Wght	2000	...	...	...	SF wt2
50	Wght	2670	...	899.975	...	Clip wt
50	Wght	1000	3000	1499.96	...	Trunnion 1
50	Wght	1000	-3000	1499.96	...	Trunnion-2
50	Wght	10000	925	...	...	Misc 10%
60	Plat	4440.85	271	3000	...	PL-3
60	Insl	2819.48	...	1346.5	...	Ins: 50
60	Nozl	1855.16	1500	2719.29	...	M1_600
60	Nozl	52.5228	2500	1992.65	...	K3_50
60	Nozl	650.85	-750	219.296	1299.04	K1_50
60	Nozl	792.68	...	3174.5	...	N1_750
60	Nozl	645.661	-750	-130.704	-1299.04	K2_50
60	Nozl	74.176	-2000	2787.84	...	K6_75
60	Wght	300	...	2675	...	N1_Vane distrib
60	Wght	100	1500	2400	...	M1_Davit
60	Wght	2000	...	-0.026015	...	SF wt-1
60	Wght	350	-750	2400	-1299.04	K2 Internals
60	Wght	350	-750	2400	1299.04	K1 Internals
60	Wght	29000	...	...	...	Inert Balls
60	Wght	850	...	2675	...	N1 insert
60	Forc	...	...	2693	...	N2_F/M

Total Weight of Each Detail Type:

Platforms	12070.8
Packing	73322.4
Insulation	7387.7
Nozzles	8254.2
Weights	128370.0

Sum of the Detail Weights 229405.1 kg.

Weight Summation Results: (kg.)

Fabricated	Shop Test	Shipping	Erected	Empty	Operating
------------	-----------	----------	---------	-------	-----------



Main Elements	81573.7	81573.7	81573.7	81573.7	81573.7	81573.7
Nozzles	8254.2	8254.2	8254.2	8254.2	8254.2	8254.2
Wld Weights	21370.0	21370.0	21370.0	21370.0	21370.0	21370.0
Platforms	...	...	12070.8	12070.8	12070.8	12070.8
Packing	...	...	73322.4	73322.4	73322.4	73322.4
Insulation	...	...	7387.7	7387.7	7387.7	7387.7
Empty Weights	...	...	...	1000.0	1000.0	...
Ope Weights	...	...	...	...	...	107000.0
Test Liquid	...	123891.0	...	...	...	...
Totals	111197.9	235088.9	203978.8	204978.8	204978.8	310978.8

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	111197.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	235088.9 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	203978.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	204978.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	204978.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	310978.8 kg.
Field Test Wt.	- Empty weight + test medium (full)	254547.5 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	285053.9 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	303308.5 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	182014.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	206440	204631
20	30	516101	511577
30	40	172034	170526
40	50	476203	457705
50	60	322098	310939
60	70	476203	457705

Total	2169079.2	2113082.5 cm ²
Total	2334.8	2274.5 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	7695.11	7695.11	7695.11	...	7695.11
20	30	11134	11134	11134	3712.57	11134
30	40	3186.01	3186.01	3186.01	...	3186.01
40	50	26773.1	103773	67933.5	4285.21	103773
50	60	121367	121367	89615	9311.6	121367
60	70	34823.4	63823.4	74983.8	2621.83	63823.4

Empty Support Force + the Sum of the Y forces	212546.81	Kgf
Operating Support Force + the Sum of the Y forces	318546.84	Kgf
Hydro Support Force + the Sum of the Y forces	262115.45	Kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
------	----	---	---------------------------------	----------------------------------



10	20	310979	310979	254547
20	30	303284	303284	246852
30	40	292150	292150	235718
40	50	288964	288964	232532
50	60	185191	185191	164599
60	70	63823.4	63823.4	74983.8

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	19931.2	19931.2	19931.2
20	30	19931.2	19931.2	19931.2
30	40	16218.6	16218.6	16218.6
40	50	16218.6	16218.6	16218.6
50	60	11933.4	11933.4	11933.4
60	70	2621.83	2621.83	2621.83

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N2_600	7073.3	10340.0	454	600	GR 1.9	0.2	7073.3	7073.3	7073.3
U1_600	7073.3	10340.0	454	600	GR 1.9	...	...	...	...
K5_75	7073.3	10340.0	454	600	GR 1.9	...	...	...	...
K4_50	7073.3	10340.0	454	600	GR 1.9	...	...	...	...
M1_600	7073.3	10340.0	454	600	GR 1.9	...	...	...	...
K3_50	7073.3	10340.0	454	600	GR 1.9	...	...	...	...
K1_50	7073.3	10340.0	454	600	GR 1.9	...	...	...	...
N1_750	7073.3	10340.0	454	600	GR 1.9	1695.0	6297.8	5378.3	7073.3
K2_50	7073.3	10340.0	454	600	GR 1.9	...	...	...	...
K6_75	7073.3	10340.0	454	600	GR 1.9	...	...	...	...
Min Rating	6297.8	10340.0, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N2_600	657.225	0.50	1	0	4658.5	7073.3	Pass
N1_750	803.402	0.13	16218	6880	4658.5	6297.8	Pass

Evaluating Flange on Nozzle [N2600]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1) + 4 \cdot \text{abs}(0) \cdot 657.225 \leq \pi \cdot 657.225^3 \cdot ((7073.296 - 4658.502) + .500 \cdot 7073.296)$$

$$= 0.16929E+02 \leq 0.54128E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2600 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7073.296, 0.500 \cdot 7073.296 + 7073.296 - 16.9 / (\pi \cdot 657.225^3))$$

$$= \min(7073.296, 10609.758)$$

$$= 7073.296 \text{ KPa.}$$

Evaluating Flange on Nozzle [N1750]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(16218) + 4 \cdot \text{abs}(6880) \cdot 803.402 \leq \pi \cdot 803.402^3 \cdot ((7073.296 - 4658.502) + .130 \cdot 7073.296)$$

$$= 0.28160E+06 \leq 0.55394E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1750 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7073.296, 0.130 \cdot 7073.296 + 7073.296 - 281601.3 / (\pi \cdot 803.402^3))$$

$$= \min(7073.296, 6297.781)$$

$$= 6297.781 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 11.7023 Hz.

The Natural Frequency for the Vessel (Ope...) is 11.7023 Hz.

The Natural Frequency for the Vessel (Filled) is 12.9418 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	7731.51	38661.7

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	7731.51	38661.7

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
60	2693.00	5467.	-7568.	5467.	12615.		12615.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.935) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.891)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	11.702 Hz
Natural Frequency of Vessel (Empty)	11.702 Hz
Natural Frequency of Vessel (Test)	12.942 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.515
Structure Height to Diameter ratio	1.886
Height to top of Structure	9947.999 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 600.0 + 300.0 = 900.0$ mm
 $= 2.953$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
Because z (2.953 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15 / 900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.515 \cdot 86.293$$

$$= 604.1 \text{ lbs. } [274.0] \text{ Kg}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt-3	900.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt-2	3000.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt-1	5000.0	0.000	0.000	0.000	0.865	1.000	78.120
Bottom head	5500.0	0.000	0.000	0.000	0.883	1.000	78.120
Shell	6425.0	0.000	0.000	0.000	0.912	1.000	78.120
Top head	8579.9	0.000	0.000	0.000	0.969	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kg
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
- Z - Maximum amplitude of vibration at top of vessel, mm
- D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/7050.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[18.28/23.13^2] \cdot 10^4 = 341.705$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.400E-06 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 307020 / (7050.0 \cdot 5615.603^2)$$

$$= 0.13810E-05$$

Since Vp is > 0.400E-06 no further vibration analysis is required !

Platform Load Calculations



ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf	Cf
PL-3	25160.00	2700.00	78.12	196.55	1.00
PL-1	42192.00	5500.00	78.12	329.60	1.00
PL-3	70320.01	10349.97	78.12	549.34	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Tailing lug	0.00	300.00	78.12	0.00
Closure plate	0.00	2806.97	78.12	0.00
Inert Balls	0.00	5499.97	78.12	0.00
SF wt2	0.00	5500.00	78.12	0.00
Clip wt	0.00	6399.97	78.12	0.00
Trunnion 1	0.00	6999.96	78.12	0.00
Trunnion-2	0.00	6999.96	78.12	0.00
Misc 10%	0.00	5500.00	78.12	0.00
N1Vane distr	0.00	10024.97	78.12	0.00
M1Davit	0.00	9749.97	78.12	0.00
SF wt-1	0.00	7349.94	78.12	0.00
K2Internals	0.00	9749.97	78.12	0.00
K1Internals	0.00	9749.97	78.12	0.00
Inert Balls	0.00	7349.97	78.12	0.00
N1 insert	0.00	10024.97	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 11.7023 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	900	6680.72	80168.7	78.12	301.431
20	30	3000	6680.72	200422	78.12	948.163
30	40	5000	6680.72	66807.2	78.12	251.193
40	50	5499.98	7071.12	2.15528	78.12	0.0081038
50	60	6425	7127.24	131854	78.12	806.133
60	70	8579.93	7071.12	160947	78.12	1334.09

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= 0.020(31.98490.75)
= 0.269 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/11.702)
= 0.085

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of (1.7 • 0.269, 0.085) per 12.8.2
= 0.085

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= 0.157/(2.0/2.0)
= 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= 0.104/(0.085(2.0/2.0))
= 1.217

Minimum value of C_s per equation 15.4-1 [Cs]:
= max((0.044 • SDS • I_e), 0.030)
= max((0.044 • 0.1568 • 2.0), 0.030)
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= 0.1568 • 310978.8
= 48761.480 Kgf

Vertical load per 12.4-4, [Ev]:
= 0.2 • Sds • W
= 0.2 • 0.157 • 310979



= 9752.3 Kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 11.7023 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	600	7695.11	126.296	125.999	2184.29
20	30	2700	11134	822.314	820.381	1942.98
30	40	4700	3186.01	409.608	408.645	1593.82
40	50	5199.98	103773	14760.8	3799.28	1493.9
50	60	6125	121367	20334.4	20286.6	-1760.42
60	70	7049.98	63823.4	12308.1	6699.78	-5566.5



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas Kgf	Cumulative Shr Eqk Cas Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	600	7731.51	7731.51	93169.6	93169.6
20	30	2700	7731.51	7731.51	83891.6	83891.6
30	40	4700	7731.51	7731.51	60696.7	60696.7
40	50	5199.98	7731.51	7731.51	52965.2	52965.2
50	60	6125	7731.51	7731.51	52965	52965
60	70	7049.98	7731.51	7731.51	38661.7	38661.7

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	600	3641.02	48761.5	19905.7	307441
20	30	2700	3339.59	48635.2	15717.2	249002
30	40	4700	2391.42	47812.9	7120.55	104327
40	50	5199.98	2140.23	47403.3	4854.74	56719.7
50	60	6125	2140.22	32642.5	4854.68	56718.5
60	70	7049.98	1334.09	12308.1	1640.92	15138.8

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	7731.51	0.00054293	0.0021351	0.34974E-05
20	30	7731.51	0.010345	0.02391	0.10637E-04
30	40	7731.51	0.029471	0.035497	0.12495E-04
40	50	7731.51	0.035497	0.035498	0.12495E-04
50	60	7731.51	0.047233	0.059288	0.13188E-04
60	70	7731.51	0.059288	0.059288	0.13188E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 35.250 Actual deflection : 0.059 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	3641.02	0.00011408	0.00043954	0.70489E-06
20	30	3339.59	0.0020211	0.0044295	0.18131E-05
30	40	2391.42	0.0053682	0.0063638	0.20411E-05
40	50	2140.23	0.0063639	0.0063639	0.20411E-05
50	60	2140.22	0.0082692	0.0102	0.20963E-05
60	70	1334.09	0.0102	0.0102	0.20963E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	1403.44	8771.5
20	30	1403.44	8771.5
30	40	1403.44	8771.5
40	50	1485.45	9284.08
50	60	1497.24	9357.77
60	70	1485.45	9284.08

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 35.250 Actual deflection : 0.010 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 35.250 Actual deflection : 0.069 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	4110.72	3836.69	557846784	520809344
20	30	4110.72	3836.69	557846784	520809344
30	40	4110.72	3836.69	557846784	520809344
40	50	8384.98	7875.96	1131736192	1064193536
50	60	16424.7	15920.2	2.198E+09	2132961280
60	70	8384.98	7875.96	1131736192	1064193536

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	115.918	196.556	-103.193	-103.193
20	30	115.918	196.556	-103.193	-103.193
30	40	82.3752	167.397	-54.9442	-122.726
40	50	164.75	334.794	-65.3073	-137.103
50	60	164.75	334.794	-74.6741	-142.208
60	70	164.75	334.794	-65.3073	-137.103

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	134.8	-3.11534	188.904
50	60	65.6121	-1.56711	91.9464
60	70	134.8	-3.11534	188.904

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-7.94883	-7.94883	-6.50641	0.3753	0.3753
20	30	-7.75214	-7.75214	-6.30971	0.3753	0.3753
30	40	-7.46755	-7.46755	-6.02512	0.30539	0.30539
40	50	-3.59806	-3.59806	-3.59806	0.14946	0.14946
50	60	-1.14077	-1.14077	-1.14077	0.054866	0.054866
60	70	-0.7947	-0.7947	-0.7947	0.02416	0.02416

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.3753	0.37482	5.789	0.18741	...
20	30	0.3753	0.29595	4.68862	0.14797	...
30	40	0.30539	0.13408	1.96444	0.067039	...
40	50	0.14946	0.044737	0.52268	0.022368	...
50	60	0.054866	0.02232	0.26077	0.01116	...
60	70	0.02416	0.015121	0.13951	0.0075606	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	3.8356
20	30	...	...	...	3.11078
30	40	...	...	...	1.32544
40	50	...	...	...	0.36309
50	60	...	...	...	0.18115
60	70	...	...	...	0.075938

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
------	----	--	---



10	20	-0.19344	0.055832
20	30	-0.19344	0.049664
30	40	-0.19344	0.040739
40	50	-0.094234	0.018601
50	60	-0.046619	-0.010844
60	70	-0.094234	-0.069312

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	1.75435	1.75435	1.63787	1.63787
20	30	1.57965	1.57965	1.47477	1.47477
30	40	1.14289	1.14289	1.06701	1.06701
40	50	0.48808	0.48808	0.45895	0.45895
50	60	0.24352	0.24352	0.23629	0.23629
60	70	0.35627	0.35627	0.33501	0.33501

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-10.65	103.19		0.1032
20		115.92	-10.20	103.19		0.0988
30		82.38	-9.24	54.94		0.1682
40		164.75	-4.37	65.31		0.0670
50		164.75	-1.51	74.67		0.0202
60		164.75	-1.28	65.31		0.0197

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-13.86	103.19		0.1343
20		115.92	-12.77	103.19		0.1237
30		82.38	-10.20	54.94		0.1857
40		164.75	-4.58	65.31		0.0701
50		164.75	-1.63	74.67		0.0218
60		164.75	-1.32	65.31		0.0202

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-10.65	103.19		0.1032
20		115.92	-10.20	103.19		0.0988



30	82.38	-9.24	54.94	0.1682
40	164.75	-4.37	65.31	0.0670
50	164.75	-1.51	74.67	0.0202
60	164.75	-1.28	65.31	0.0197

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.03	115.92	-15.81	103.19	0.0002	0.1532
20		115.92	-14.35	103.19		0.1390
30		82.38	-10.84	54.94		0.1973
40		164.75	-4.74	65.31		0.0726
50		164.75	-1.71	74.67		0.0229
60		164.75	-1.38	65.31		0.0212

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-7.07	103.19		0.0685
20		196.56	-6.83	103.19		0.0662
30		167.40	-6.40	122.73		0.0521
40		334.79	-3.77	137.10		0.0275
50		334.79	-1.21	142.21		0.0085
60		334.79	-0.83	137.10		0.0060

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-6.88	103.19		0.0667
20		196.56	-6.69	103.19		0.0648
30		167.40	-6.33	122.73		0.0516
40		334.79	-3.75	137.10		0.0273
50		334.79	-1.20	142.21		0.0084
60		334.79	-0.82	137.10		0.0060

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-10.65	103.19		0.1032
20		115.92	-10.20	103.19		0.0988
30		82.38	-9.24	54.94		0.1682
40	131.79	164.75		65.31	0.7999	
50	64.75	164.75		74.67	0.3930	
60	134.31	164.75		65.31	0.8152	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.03	115.92	-15.81	103.19	0.0002	0.1532
20		115.92	-14.35	103.19		0.1390
30		82.38	-10.84	54.94		0.1973
40	132.38	164.75		65.31	0.8035	
50	65.02	164.75		74.67	0.3947	
60	134.46	164.75		65.31	0.8161	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-10.65	103.19		0.1032
20		115.92	-10.20	103.19		0.0988
30		82.38	-9.24	54.94		0.1682
40		164.75	-7.49	65.31		0.1147
50		164.75	-3.08	74.67		0.0412
60		164.75	-4.40	65.31		0.0674

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.03	115.92	-15.81	103.19	0.0002	0.1532
20		115.92	-14.35	103.19		0.1390
30		82.38	-10.84	54.94		0.1973
40		164.75	-7.86	65.31		0.1203
50		164.75	-3.28	74.67		0.0439
60		164.75	-4.50	65.31		0.0689

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-7.07	103.19		0.0685
20		196.56	-6.83	103.19		0.0662
30		167.40	-6.40	122.73		0.0521
40	185.48	334.79		137.10	0.5540	
50	90.87	334.79		142.21	0.2714	
60	188.14	334.79		137.10	0.5620	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-6.88	103.19		0.0667
20		196.56	-6.69	103.19		0.0648
30		167.40	-6.33	122.73		0.0516
40	185.46	334.79		137.10	0.5539	
50	90.86	334.79		142.21	0.2714	
60	188.13	334.79		137.10	0.5619	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-8.32	103.19		0.0807
20		115.92	-8.13	103.19		0.0788
30		82.38	-7.77	54.94		0.1415
40	131.35	164.75		65.31	0.7973	
50	64.53	164.75		74.67	0.3917	
60	134.03	164.75		65.31	0.8135	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.42	103.19		0.0719
20		115.92	-7.24	103.19		0.0701
30		82.38	-6.97	54.94		0.1269
40	131.42	164.75		65.31	0.7977	
50	64.51	164.75		74.67	0.3915	
60	134.05	164.75		65.31	0.8137	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-8.32	103.19		0.0807
20		115.92	-8.13	103.19		0.0788
30		82.38	-7.77	54.94		0.1415
40	131.35	164.75		65.31	0.7973	
50	64.53	164.75		74.67	0.3917	
60	134.03	164.75		65.31	0.8135	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		115.92	-8.32	103.19		0.0807
20		115.92	-8.13	103.19		0.0788
30		82.38	-7.77	54.94		0.1415
40	131.35	164.75		65.31	0.7973	
50	64.53	164.75		74.67	0.3917	
60	134.03	164.75		65.31	0.8135	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-8.32	103.19		0.0807
20		115.92	-8.13	103.19		0.0788
30		82.38	-7.77	54.94		0.1415
40		164.75	-3.75	65.31		0.0574
50		164.75	-1.20	74.67		0.0160
60		164.75	-0.82	65.31		0.0125

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.60	-10.02	85.99		0.1166
20		96.60	-9.66	85.99		0.1123
30		68.65	-8.88	45.79		0.1938
40	131.86	137.29		54.42	0.9604	
50	64.76	137.29		62.23	0.4717	
60	134.32	137.29		54.42	0.9783	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.60	-10.02	85.99		0.1166
20		96.60	-9.66	85.99		0.1123
30		68.65	-8.88	45.79		0.1938
40		137.29	-7.33	54.42		0.1347
50		137.29	-3.02	62.23		0.0485
60		137.29	-4.36	54.42		0.0801

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-8.32	103.19		0.0807
20		115.92	-8.13	103.19		0.0788
30		82.38	-7.77	54.94		0.1415
40	131.35	164.75		65.31	0.7973	
50	64.53	164.75		74.67	0.3917	
60	134.03	164.75		65.31	0.8135	

Absolute Maximum of the all of the Stress Ratio's

0.9783

Governing Element: Top head

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-3	...	-7.9	0.4	1.8	...	+ Tens
Skirt-3	...	-7.9	-0.4	-1.8	-10.0	- Comp
Skirt-2	...	-7.7	0.4	1.6	...	+ Tens
Skirt-2	...	-7.7	-0.4	-1.6	-9.7	- Comp
Skirt-1	...	-7.4	0.3	1.1	...	+ Tens
Skirt-1	...	-7.4	-0.3	-1.1	-8.9	- Comp



Bottom head	134.8	-3.6	0.1	0.5	131.9	+ Tens
Bottom head	134.8	-3.6	-0.1	-0.5	...	- Comp
Shell	65.6	-1.2	0.1	0.2	64.8	+ Tens
Shell	65.6	-1.2	-0.1	-0.2	...	- Comp
Top head	134.8	-0.9	...	0.4	134.3	+ Tens
Top head	134.8	-0.9	...	-0.4	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

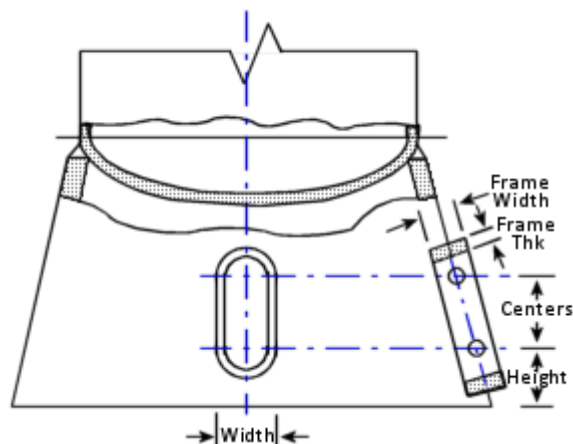
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	292039.688 Kgf
Weight of vessel contents (liquid)	Ff	0.000 Kgf
Vessel Gross Weight	Fg + Ff	292039.688 Kgf
Skirt Mean Diameter	Dz	5452.000 mm
Skirt Analysis Thickness (Corroded)	ez	22.400 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	-7568.000 Kgf
Moment at section 1-1	M1	165023.984 Kg-m.
Applied Moment - Bottom of Skirt	Mb	332893.906 Kg-m.
Weight at Bottom of Skirt	W	303283.688 Kgf



Access Opening: 1 Cuts through the top of the skirt.
 Access Opening: 2 Cuts through the bottom of the skirt.
 Access Opening: 3 Cuts through the top of the skirt.
 Access Opening: 4 Cuts through the top of the skirt.
 Access Opening: 6 Cuts through the top of the skirt.

Note:

If an opening cuts the skirt top or bottom, this is not a problem.
 This becomes a 'Sentry Box' or 'Mouse Hole' type Opening.

Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	50.8000	0.0	3000.00	0.00	80.000	16.000
2	800.0000	45.0	300.00	0.00	80.000	16.000
3	50.8000	90.0	3000.00	0.00	80.000	16.000
4	50.8000	180.0	3000.00	0.00	80.000	16.000
5	800.0000	180.0	1100.00	0.00	80.000	16.000
6	50.8000	270.0	3000.00	0.00	80.000	16.000

**Interference check:**

Access	Access	Pitch mm	Ligament mm
1	2	3445.8245	3020.4246
2	3	3445.8245	3020.4246
3	4	4281.9194	4231.1191
4	5	1899.9999	1474.5999
5	6	4684.5312	4259.1309
1	6	4281.9194	4231.1191

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 1899.9999 mm

Minimum Ligament between Openings is: 1474.5999 mm

Recommended minimum Ligament [lMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(5452.0 \cdot 22.4)} \text{ mm}$$

$$= 698.927 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 2

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
2	36.5622	53.4378

The points for integrating lxx, lyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
0.000	36.562
-306.562	0.000

Height of Weakest Section	H	299.404	mm
Rotation of Principle Axes	a	-40.153	deg.
Centroid Position in X direction	g	-61.171	mm
Centroid Position in Y direction	h	-79.991	mm
Principle Moment of Inertia	Ixx	131642480.000	cm ⁴
Principle Moment of Inertia	Iyy	143525312.000	cm ⁴
Distance to Extreme Fiber	ybar	2837.784	mm
Distance to Extreme Fiber	xbar	2742.026	mm
Minimum Section Modulus	W4	463892160.000	mm ³
Metal Area at the Cutting Plane	A4	3708.015	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(800.0/5452.0)$$

$$= 0.1473 \text{ rad (8.4378 deg)}$$

Note: The largest opening width is considered.

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 5452.0/2$$

$$= 2726.0 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
25.6	7.8	...	...



25.6	176.2	...	...
25.6	7.8	...	...
25.6	7.8	...	...
25.6	176.2	...	...
25.6	7.8	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.147 \cdot \sqrt{(2726.0/22.4)}$$

$$= 1.625$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= ex(\text{new}) - ci - c_{ext}$$

$$= 24.0 - 0.8 - 0.8$$

$$= 22.4 \text{ mm}$$

Note: At least one of the openings cuts the Top Of the Skirt (Ok)

Note: At least one of the openings cuts the Bottom Of the Skirt (Ok)

Find the moment acting at the Critical Height [M4]:

$$= (M_{bottom} - M_{top}) \cdot (H - \text{CutHeight}) / H + M_{top}$$

$$= (332893.906 - 165023.984) \cdot (3000.0 - 299.404) / 3000.0 + 165023.984$$

$$= 316140.281 \text{ Kg-m.}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{bar} = 0.13164E+09 / 2837.784 = 0.46389E+09 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{bar} = 0.14353E+09 / 2742.026 = 0.52343E+09 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(.46389E+09, .52343E+09) = 0.46389E+09 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))}$$

$$= \sqrt{(\min(0.13164E+09, 0.14353E+09) / 3708.015)}$$

$$= 1884.202 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 1884.202 / 3000.0$$

$$= 0.628$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(-61.171^2 + -79.991^2)}$$

$$= 100.700 \text{ mm}$$

Where $e_x = -61.171$: $e_y = -79.991$

Moment Increment at Centroid [ΔM_4]:

$$= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$$

$$= \text{abs}(-7568.0 - 292039.69 - 0.0) \cdot 100.7$$

$$= 30171.096 \text{ Kg-m.}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$= \min(1; A_4 / (\pi \cdot D_3 \cdot e_{a3}))$$

$$= \min(1; 3708.015 / (\pi \cdot 5452.0 \cdot 22.4))$$

$$= 0.966$$

Weakening Factor, 16.12-70(c) [psi2]:



$$\begin{aligned} &= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / (\pi \cdot D3 \cdot ea3)) \\ &= \min(1; 4.0 \cdot 0.52343E+09 / (\pi \cdot 5452.0^2 \cdot 22.4)) \\ &= 1.000 \end{aligned}$$

Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 5452.0^2 \cdot 22.4 \cdot 103.2 \\ &= 5502759.0 \text{ Kg-m.} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call} \\ &= \pi \cdot 5452.0 \cdot 22.4 \cdot 103.2 \\ &= 4037161.0 \text{ Kg} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F4 / (\psi1 \cdot F_{c,max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(yq)) / (\psi2 \cdot M_{max}) \\ &= 299607.7 / (0.966 \cdot 4037161) + (\text{abs}(316140.3) + 299607.7 \cdot \text{abs}(100.7)) / \\ &\quad (1.0 \cdot 5502759) \\ &= 0.140 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

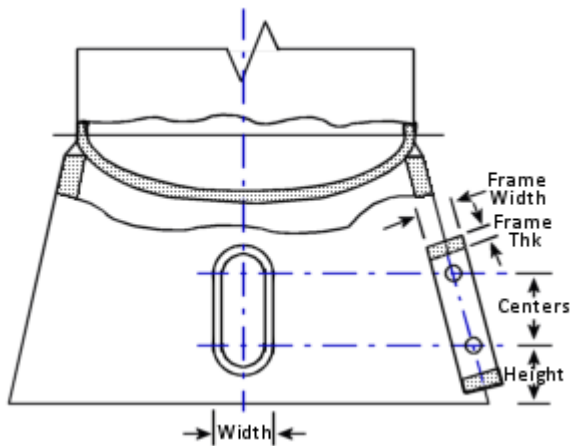
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

Total vessel (empty) weight	Fg	292039.688 Kg
Head Weight below section 2-2	DeltaFg	103773.078 Kg
Weight of vessel contents (liquid)	Ff	0.000 Kg
Vessel Gross Weight	Fg + Ff	292039.688 Kg
Skirt Mean Diameter	Dz	5452.000 mm
Skirt Analysis Thickness (Corroded)	ez	22.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-7568.000 Kg
Moment at section 1-1	M1	109684.961 Kg-m.
Applied Moment - Bottom of Skirt	Mb	165023.984 Kg-m.
Weight at Bottom of Skirt	W	292149.719 Kg

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	5450.000 mm
Head Corroded Thickness	eb	46.000 mm



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	142552576.000 cm ⁴
Moment of Inertia	Iyy =	142552576.000 cm ⁴
Section Modulus	W =	522937280.000 mm ³
Section Metal Area	A =	3836.664 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\frac{\min(Ixx, Iyy)}{Area}}$$

$$= \sqrt{\frac{\min(0.14255E+09, 0.14255E+09)}{3836.664}}$$

$$= 1927.573 \text{ mm}$$

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 1927.573/1000.0$$

$$= 1.928$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 109684.961 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -109684.961 / .20917E+10 - 292149.719 / 3836.66431 \\ &= -7.982 \text{ N./mm}^2 \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +109684.961 / .20917E+10 - 292149.719 / 3836.66431 \\ &= -6.953 \text{ N./mm}^2 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F1 - Fg - Ff + 4 \cdot M1 / Dz \\ &= -7568.0 - 292039.688 - 0.0 + 4 \cdot 109684.961 / 5452.0 \\ &= -219136.1 \text{ Kg} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F1 - Fg - Ff - 4 \cdot M1 / Dz \\ &= -7568.0 - 292039.688 - 0.0 - 4 \cdot 109684.961 / 5452.0 \\ &= -380079.3 \text{ Kg} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (-0.2E+06 + 103773.078 + 0.0) / (\pi \cdot 5450.0 \cdot 46.0) + 4621.0 \cdot 5450.0 / (4 \cdot 46.0) \\ &= 135.44 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (-0.4E+06 + 103773.078 + 0.0) / (\pi \cdot 5450.0 \cdot 46.0) + 4621.0 \cdot 5450.0 / (4 \cdot 46.0) \\ &= 133.44 \text{ N./mm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (0.0 + 103773.078) / (\pi \cdot 5450.0 \cdot 46.0) + 4621.0 \cdot 5450.0 / (4 \cdot 46.0) \\ &= 138.17 \text{ N./mm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= Fzp / (\pi \cdot Dz \cdot ez) \\ &= -219136.094 / (\pi \cdot 5452.0 \cdot 22.4) \\ &= -5.60 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= Fzq / (\pi \cdot Dz \cdot ez) \\ &= -380079.312 / (\pi \cdot 5452.0 \cdot 22.4) \\ &= -9.72 \text{ N./mm}^2 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$\begin{aligned} &= 1 - 0.5(Db + eb - Dz + ez) / (r + eb) \\ &= 1 - 0.5(5450.0 + 46.0 - 5452.0 + 22.4) / (2702.0 + 46.0) \\ &= 0.9879 \end{aligned}$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.988) = 8.915 \text{ deg}$$

Dimension [a]:

$$\begin{aligned} &= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ &= 0.5 \cdot \sqrt{(46.0^2 + 22.4^2 + 2 \cdot 46.0 \cdot 22.4 \cdot 0.988)} \\ &= 34.109 \text{ mm} \end{aligned}$$



Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 34.10 \cdot -219136.1 = -7474.63 \text{ Kg-m.} \\ M_q &= a \cdot F_{zq} = 34.10 \cdot -380079.3 = -12964.34 \text{ Kg-m.} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{e_b}{e_z} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{46.0}{22.4} \right)^2 \\ &= 0.390 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.39 \cdot 6 \cdot -7474.634 / (\pi \cdot 5450.0 \cdot 46.0^2) \\ &= -4.73 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.39 \cdot 6 \cdot -12964.336 / (\pi \cdot 5450.0 \cdot 46.0^2) \\ &= -8.20 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.39 \cdot 6 \cdot -7474.634 / (\pi \cdot 5452.0 \cdot 22.4^2) \\ &= -19.94 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.39 \cdot 6 \cdot -12964.336 / (\pi \cdot 5452.0 \cdot 22.4^2) \\ &= -34.58 \text{ N./mm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} e_b/D_b &= 1.811/214.567 = 0.0084 \\ y &= 125 \cdot e_b/D_b = 125 \cdot 46.0/5450.0 = 1.055 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ N./mm}^2 \text{ (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 135.443 - -4.73 + 0.0 \\ &= 140.17 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 135.443 + -4.73 - 0.0 \\ &= 130.71 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 133.439 - -8.204 + 0.0 \\ &= 141.64 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 133.439 + -8.204 - 0.0 \\ &= 125.24 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 138.172 + -4.73 + 0.0 \\ &= 133.44 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2potot}]:



$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 138.172 - -4.73 - 0.0 \\
 &= 142.90 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qitot}]: \\
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 138.172 + -8.204 + 0.0 \\
 &= 129.97 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qotot}]: \\
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 138.172 - -8.204 - 0.0 \\
 &= 146.38 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3pitot}]: \\
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= -5.601 - -19.939 \\
 &= 14.34 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3potot}]: \\
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= -5.601 + -19.939 \\
 &= -25.54 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qitot}]: \\
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -9.715 - -34.583 \\
 &= 24.87 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}]: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -9.715 + -34.583 \\
 &= -44.30 \text{ N./mm}^2
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 137.292(3 - (1/1.5)) \cdot (135.443/137.292)^2 \\
 &= 322.80 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 137.292(3 - (1/1.5)) \cdot (133.439/137.292)^2 \\
 &= 325.41 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 137.292(3 - (1/1.5)) \cdot (138.172/137.292)^2 \\
 &= 319.17 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 137.292(3 - (1/1.5)) \cdot (138.172/137.292)^2 \\
 &= 319.17 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 137.292(3 - (1/1.5)) \cdot (-5.601/137.292)^2 \\
 &= 411.72 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 137.292(3 - (1/1.5)) \cdot (-9.715/137.292)^2 \\
 &= 411.42 \text{ N./mm}^2
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)



Stress	Actual	Allowed	Result	Equation
Stot1pi	140.17	322.80	Pass	46
Stot1po	130.71	322.80	Pass	47
Stot1qi	141.64	325.41	Pass	48
Stot1qo	125.24	325.41	Pass	49
Stot2pi	133.44	319.17	Pass	50
Stot2po	142.90	319.17	Pass	51
Stot2qi	129.97	319.17	Pass	52
Stot2qo	146.38	319.17	Pass	53
Stot3pi	14.34	411.72	Pass	54
Stot3po	-25.54	411.72	Pass	55
Stot3qi	24.87	411.42	Pass	56
Stot3qo	-44.30	411.42	Pass	57

The stresses are satisfactory in the region without openings.

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**Shop/Field Installation Options :**

Platform(s) installed in the Shop.

Packing is installed in the Shop.

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	6618.739 mm
Center of Gravity of the Packing	6125.000 mm
Center of Gravity of the Insulation	6128.412 mm
Center of Gravity of the Nozzles	5763.785 mm
Center of Gravity of the Added Weights (Operating)	5689.834 mm
Center of Gravity of the Added Weights (Empty)	5612.747 mm
Center of Gravity of Bare Shell New and Cold	5248.431 mm
Center of Gravity of Bare Shell Corroded	5253.884 mm
Vessel CG in the Operating Condition	5732.614 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	5734.908 mm
Vessel CG in the Test Condition	5881.859 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	5476.0000	mm
Skirt Thickness	STHK	24.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	114.00	N./mm ²
Basing Yield Stress		248.00	N./mm ²
Inside Diameter of Basing	DI, D4	5236.0000	mm
Outside Diameter of Basing	DOU, D5	5756.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm ²
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	5616.0000	mm
Bolt Allowable Shear Stress		82.800	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	130.0000	mm
Width of Gussets at Base Plate	BWDT	130.0000	mm
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	248.0	N./mm ²
Height of Gussets	HG, h1	268.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm
External Corrosion Allowance	CA	3.0000	mm
Dead Weight of Vessel	DW	212546.8	Kgf
Operating Weight of Vessel	ROW	318546.8	Kgf
Earthquake Moment on Basing	EQMOM	316316.6	Kg-m.
Wind Moment on Basing	WIMOM	28780.9	Kg-m.
User Moment at the Base [Earthquake Case]		93169.6	Kg-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		93169.6	Kg-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).



Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 409486/5616.0) - 308795) / 24$$

$$= -714.3 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 121950/5616.0) - 212547) / 24$$

$$= -5237.1 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/5616.0) - 318547) / 24$$

$$= -13272.8 \text{ Kgf}$$

Governing Bolt Load Condition, **** No Tensile Bolt Loads ****

Area Available in 1 Bolt (corroded)	Abss :	11.5517	cm ²
Area Available in all the Bolts	Abss * RN :	277.2397	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.150	1.511	2.441	0.438	0.781	93335.7	93335.7	5590.6
4	0.075	0.113	0.737	2.946	0.485	0.763	95531.3	95531.3	4741.2
6	0.132	0.141	0.980	2.814	0.473	0.770	94735.6	94735.6	4921.8
8	0.136	0.139	0.998	2.804	0.472	0.770	94679.2	94679.2	4937.7
10	0.137	0.137	1.003	2.801	0.472	0.770	94665.3	94665.3	4941.7
12	0.136	0.137	0.999	2.803	0.472	0.770	94675.7	94675.7	4938.7

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 94675.7 / (1.571 \cdot 5616.0 \cdot 2.803)$$

$$= 75.071 \text{ N./mm}^2, \text{ Should be } \leq 138.0 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 24 \cdot 11.552 / (\pi \cdot 5616.0)$$

$$= 1.571 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 1.208[(2 \cdot 0.1365 \cdot 5616.0 + 260.0) / (2 \cdot 0.1365 \cdot 5616.0)]$$

$$= 1.413 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 140.0, b = 367.566, l/b = 0.380884

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(271.268, 1120.934) = 1120.934 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 1120.934 / 163.7)} + 3.0$$

$$= 23.074 \text{ mm}$$

Nomenclature:

a	= (Dc-Ds)/2	Skirt Distance to Bolt Circle
P	= Sa * Abss	Maximum Load on one Bolt
l		Radial Top Plate Width
lavg		Average Gusset Plate Width
g1	= Gamma 1	Constant Term f(b/l)
g2	= Gamma 2	Constant Term f(b/l)
g	= Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb		Allowable Bending Stress

Values for table 10.6, l = 140.0, b = 96.0, b/l = 0.685714

As b/l (0.686) is less than 1, inverting b/l = 1.458.

Moment Term, based on geometry [Mo]:



$$\begin{aligned}
 &= P/(4 \cdot \pi) [1.3(\ln((2\sin(\pi \cdot a/l)/(\pi \cdot g))) + 1)] - [(0.7 - g/2)P/(4\pi)] \\
 &= 16255.28/(4 \cdot \pi) [1.3(\ln((2 \cdot 140.0 \cdot \sin(\pi \cdot 70.0/140.0))/(\pi \cdot 40.0))) + 1] - [(0.7 - 0.076) \cdot 16255.28/(4 \cdot \pi)] \\
 &= 1833.0144 \text{ Kgf}
 \end{aligned}$$

Required Thickness of Continuous Top Ring [Tc]:

$$\begin{aligned}
 &= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}} \\
 &= \sqrt{(6 \cdot \text{Abs}(1833.01)/165.33) + 3.0} \\
 &= 28.5415 \text{ mm}
 \end{aligned}$$

Required Gusset Plate Thickness [tg]:

$$\begin{aligned}
 &= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca} \\
 &= 16255.28/(124.11 \cdot 130.0) + 3.0 \\
 &= 12.88 \text{ (not less than } 9.525 + 3.0 \text{) mm}
 \end{aligned}$$

Bolt spacing [m]:

$$\begin{aligned}
 &= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts} \\
 &= \pi \cdot 5616.0/24 = 735.133 \text{ mm}
 \end{aligned}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$\begin{aligned}
 &= 1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca} \\
 &= 1.76(16255 \cdot 70.0 / (735.13 \cdot 306.0 \cdot 207))^{2/3} \cdot 2738.01^{1/3} + \text{Ca} \\
 &= 9.5 + 1.6 = 11.1 \text{ mm}
 \end{aligned}$$

Shear Stress in a Single Bolt [tb]:

$$\begin{aligned}
 &= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 56493.0 / (11.552 \cdot 24) \\
 &= 20.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.
 \end{aligned}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	23.0741	mm
Actual Basing Thickness as entered by user	38.0000	mm
Required Thickness of Chair Cap	28.5415	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	12.8804	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	11.0996	mm
Given Thickness of Skirt	24.0000	mm
Required Gusset Height to meet local stress	55.1000	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$\begin{aligned}
 &= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA}) \\
 &= 15.812 \cdot (24.0 - 1.6) \\
 &= 36.12 \text{ Kgf/mm}
 \end{aligned}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$\begin{aligned}
 &= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\
 &= 36 / [(0.4 \cdot 248) \cdot 2 \cdot 0.707] \\
 &= 2.525 \text{ mm}
 \end{aligned}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$\begin{aligned}
 &= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta})) \\
 &= 16255.3 / (172.0 + 2 \cdot (268.0 + 44.0)) \\
 &= 20.421 \text{ Kgf/mm}
 \end{aligned}$$

Horizontal Plate Load [Wh]:

$$\begin{aligned}
 &= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2) \\
 &= 16255.3 \cdot 70.0 / (172.0 \cdot (312.0) + 0.6667 \cdot (312.0)^2) \\
 &= 9.597 \text{ Kgf/mm}
 \end{aligned}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$



$$= \sqrt{20.42^2 + 9.6^2}$$
$$= 22.564 \text{ Kgf/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 22.56 / [(0.4 \cdot 248) \cdot 2 \cdot 0.707]$$
$$= 1.578 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 16255.3 / (2 \cdot 130.0)$$
$$= 62.52 \text{ Kgf/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 16255.3 \cdot 70.0 / (2 \cdot 312.0 \cdot 130.0)$$
$$= 14.027 \text{ Kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$
$$= \sqrt{62.52^2 + 14.03^2}$$
$$= 64.075 \text{ Kgf/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 64.07 / [(0.4 \cdot 248) \cdot 2 \cdot 0.707]$$
$$= 4.48 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	4.4798	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2600

From: 40

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	49.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.04	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	300.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Piping Exit Angle [North Clockwise reference]		45.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		4000.0	mm
Centerline Distance to Tangent		4000.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	600.000	mm.
Actual Thickness Used in Calculation	20.000	mm



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
= 45.6284 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4621.0 \cdot 300.0 / (126 \cdot 1.0 + 0.4 \cdot 4621.0)$
= 10.8410 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.6106 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 60.0 / 0.5773
= 103.9231 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1132.0001	mm
Parallel to Vessel Wall, opening length	d	566.0001	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	115.0000	mm
Normal to Vessel Wall, Inward		35.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 126.0 / 137.3)
= 0.918

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 126.0 / 137.3)
= 0.918

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.918, 1.0)
= 0.918

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	264.018	26.370	NA
Area in Shell	A1	2.056	204.114	NA
Area in Nozzle Wall	A2	13.004	32.493	NA
Area in Inward Nozzle	A3	156.245	156.245	NA
Area in Welds	A41+A42+A43	0.918	0.918	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	108.095	108.095	NA
TOTAL AREA AVAILABLE	Atot	280.319	501.866	NA



The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (566.0•45.6284•1.0+2•77.0•45.6284•1.0•(1-0.92))
= 264.018 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 566.0(1.0 • 46.0 - 1.0 • 45.628) - 2 • 77.0
(1.0 • 46.0 - 1.0 • 45.6284) • (1 - 0.918)
= 2.056 cm²

Area Available in Nozzle Projecting Outward [A2]:
= (2 • tlnp)(tn - trn)fr2
= (2 • 115.0)(17.0 - 10.84) 0.918
= 13.004 cm²

Area Available in Inward Nozzle [A3]:
= 2 • ti • min(h, TI, 2.5 • ti) • fr2
= 2 • 74.0 • (115.0) • 0.918
= 156.245 cm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:
= Wo² • fr2 + (Wi-can/0.707)² • fr2
= 10.0² • 0.918 + (0.0)² • 0.918
= 0.918 cm²

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(115.0,300.0,70.0)) • (80.0 - 20.0) • 0.918)
= 108.095 cm² Includes Bevel Area of 30.983 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 13.8410 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 48.6284 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 48.6284 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0178 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[11.331, max(48.6284, 4.5)]
= 11.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(13.841, 11.3312)
= 13.8410 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 37.5,	Allowable	: 126.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 292.5 N./mm ²	Passed
Occasional	: 37.3,	Allowable	: 167.6 N./mm ²	Passed
Shear	: 0.4,	Allowable	: 88.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of



the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20.0, tr = 9.914, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.583, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -16 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4621.00/10340.00 = 0.447

Weld Size Calculations, Description: N2600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (264.0185 - 2.0561 + 2 • 77.0 • 0.918 •
(1.0 • 46.0 - 45.6284))137)
= 367472.94 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (121.0996 + 0.918 - 0.0 • 0.92) • 137
= 170820.12 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (121.0996 + 156.2448 + 0.918 + (65.0316)) • 137
= 480598.88 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (121.0996 + 156.2448 + 0.918 + 0.0 + (65.0316)) • 137
= 480598.88 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 600.0 • 10.0 • 0.49 • 126
= 59351. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 291.5) \cdot (80.0 - 3.0) \cdot 0.7 \cdot 126$$

$$= 634366. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 600.0 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137$$

$$= 449135. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (59351 + 634366) = 693717 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (59351 + 0 + 449135 + 0) = 508487 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (59351 + 449135 + 0) = 508487 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 693716 Kg, must exceed W = 367472 Kg or W1 = 170820 Kg
 Path 2-2 = 508486 Kg, must exceed W = 367472 Kg or W2 = 480598 Kg
 Path 3-3 = 508486 Kg, must exceed W = 367472 Kg or W3 = 480598 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 16.7246 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 648.9999 mm

Input Echo, WRC107/537 Item 1, Description: N2600 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5398.000	mm
Vessel Thickness	Tv	49.000	mm
Design Temperature	T1	454.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	137.29	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	720.000	mm
Nozzle Thickness	Tn	80.000	mm
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	126.04	N./mm ²
Design Internal Pressure	Dp	4621.000	KPa.
Include Pressure Thrust		No	

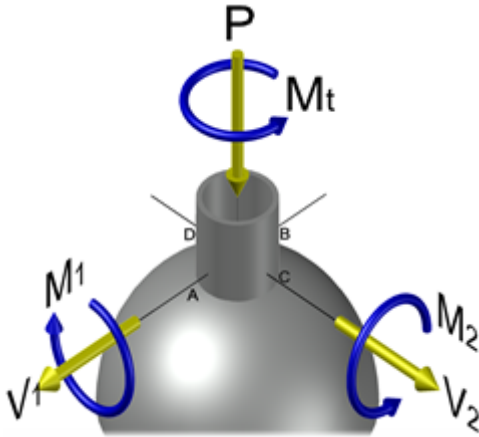
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	561.4	Kgf
Longitudinal Shear	(SUS)	(V1) V1	405.4	Kgf
Circumferential Shear	(SUS)	(Vc) V2	405.4	Kgf
Circumferential Moment	(SUS)	(Mc) M1	0.7	Kg-m.
Longitudinal Moment	(SUS)	(M1) M2	0.7	Kg-m.
Torsional Moment	(SUS)	Mt	0.9	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 720.0 + 2 \cdot 1.65 \cdot \sqrt{(2725.0 (49.0 - 3.0))}$$

$$= 1888.358 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	561.4	Kgf
Circumferential Shear	(VC) V2	405.4	Kgf
Longitudinal Shear	(VL) V1	405.4	Kgf
Circumferential Moment	(MC) M1	0.7	Kg-m.
Longitudinal Moment	(ML) M2	0.7	Kg-m.
Torsional Moment	MT	0.9	Kg-m.

Dimensionless Param: U = 1.02 TAU = 5.00 (4.18) RHO = 0.60

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.03998	(A,B,C,D)
M(x) / P	SP 1	0.04096	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.04920	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.07359	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.04920	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.07359	(A,B,C,D)
N(y) * T / P	SP 1	0.06738	(A,B,C,D)
M(y) / P	SP 1	0.01241	(A,B,C,D)

$N(y) * T * \sqrt{R_m * T} / MC$ SM 1 0.06693 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / MC$ SM 1 0.02540 (A,B,C,D)
 $N(y) * T * \sqrt{R_m * T} / ML$ SM 1 0.06693 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / ML$ SM 1 0.02540 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad. Bend. P		-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Rad. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-0.7	0.5	-0.7	0.5	-0.7	0.5	-0.7	0.5
Tang. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Tang. Bend. P		-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Tang. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Tang. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-0.4	0.0	-0.4	0.0	-0.4	0.0	-0.4	0.0
Shear VC		0.1	0.1	-0.1	-0.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		0.1	0.1	-0.1	-0.1	-0.1	-0.1	0.1	0.1
Str. Int.		0.8	0.6	0.8	0.5	0.8	0.6	0.8	0.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		135.8	135.8	135.8	135.8	135.8	135.8	135.8	135.8
Rad. Pl (SUS)		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad. Q (SUS)		-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Long. Pm (SUS)		135.8	135.8	135.8	135.8	135.8	135.8	135.8	135.8
Long. Pl (SUS)		-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Long. Q (SUS)		-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.1	0.1	-0.1	-0.1	-0.1	-0.1	0.1	0.1
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		135.8	135.8	135.8	135.8	135.8	135.8	135.8	135.8
Pm+Pl (SUS)		135.7	135.7	135.7	135.7	135.7	135.7	135.7	135.7
Pm+Pl+Q (Total)		135.4	136.3	135.4	136.3	135.4	136.3	135.4	136.3

Vessel Stress Summation Comparison (N./mm²):

Type of	Max. S.I.	S.I. Allowable	Result
---------	-----------	----------------	--------



Stress Int.			
Pm (SUS)	135.75	137.29	Passed
Pm+Pl (SUS)	135.70	205.94	Passed
Pm+Pl+Q (TOTAL)	136.30	411.88	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: U1600

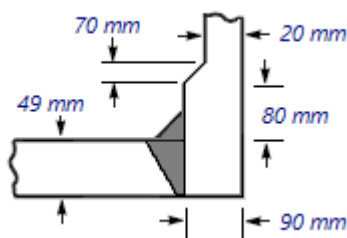
From: 40

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	49.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.04	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	70.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	1000.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		120.0000	deg.
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: U1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 600.000 mm.
Actual Thickness Used in Calculation 20.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
= 45.6284 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4621.0 \cdot 300.0 / (126 \cdot 1.0 + 0.4 \cdot 4621.0)$
= 10.8410 mm

Required Nozzle thickness under External Pressure per UG-28 : 3.1545 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 70.0/0.5773
= 121.2436 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

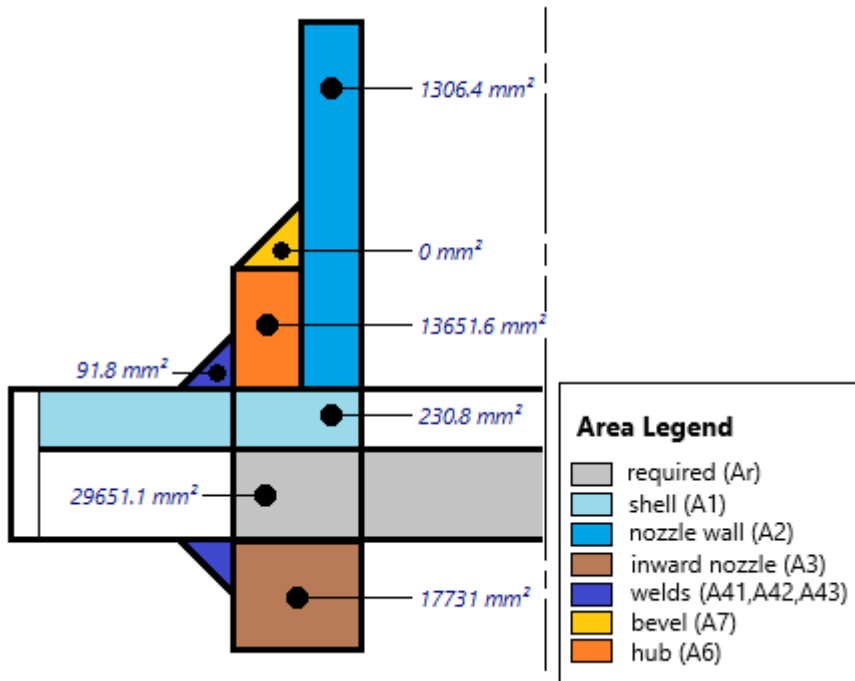
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1271.0529	mm
Parallel to Vessel Wall, opening length	d	635.5264	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	115.0000	mm
Normal to Vessel Wall, Inward		35.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 126.0/137.3)
= 0.918

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 126.0/137.3)
= 0.918

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.918, 1.0)
= 0.918



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	296.491	29.613	NA
Area in Shell	A1	2.308	229.154	NA
Area in Nozzle Wall	A2	13.074	29.390	NA
Area in Inward Nozzle	A3	177.359	177.359	NA
Area in Welds	A41+A42+A43	0.918	0.918	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	136.554	136.554	NA
TOTAL AREA AVAILABLE	Atot	330.213	573.375	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 62.95 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (635.5264 \cdot 45.6284 \cdot 1.0 + 2 \cdot 87.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 296.491 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 635.526(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 87.0$
 $(1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918)$
 $= 2.308 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 115.0)(17.0 - 10.84)0.918/\sin(95.9)$
 $= 13.074 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 84.0 \cdot (115.0) \cdot 0.918$



= 177.359 cm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2 \\ &= 10.0^2 \cdot 0.918 + (0.0)^2 \cdot 0.918 \\ &= 0.918 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(115.0, 1500.0, 80.0)) \cdot (90.0 - 20.0) \cdot 0.918 \\ &= 136.554 \text{ cm}^2 \text{ Includes Bevel Area of } 33.737 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 13.8410 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 48.6284 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 48.6284 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0178 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.331, \max(48.6284, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(13.841, 11.3312) \\ &= 13.8410 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20.0, tr = 9.914, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.583, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -16 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4621.00/10340.00 = 0.447$$

Weld Size Calculations, Description: U1600

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr))Sv)$$



$$\begin{aligned} &= \max(0, (296.4905 - 2.3084 + 2 \cdot 87.0 \cdot 0.918 \cdot \\ &\quad (1.0 \cdot 46.0 - 45.6284)) \cdot 137) \\ &= 412675.00 \text{ Kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

$$\begin{aligned} \text{Weld Load [W1]:} \\ &= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v \\ &= (149.6273 + 0.918 - 0.0 \cdot 0.92) \cdot 137 \\ &= 210757.92 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W2]:} \\ &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v \\ &= (149.6273 + 177.3589 + 0.918 + (73.4773)) \cdot 137 \\ &= 561919.31 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W3]:} \\ &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S \\ &= (149.6273 + 177.3589 + 0.918 + 0.0 + (73.4773)) \cdot 137 \\ &= 561919.31 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

$$\begin{aligned} \text{Shear, Outward Nozzle Weld [Sonw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 673.7029 \cdot 10.0 \cdot 0.49 \cdot 126 \\ &= 66642. \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Shear, Nozzle Wall [Snw]:} \\ &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 327.3073) \cdot (90.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 804795. \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension, Shell Groove Weld [Tngw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 673.7029 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 504306. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

$$\begin{aligned} \text{Failure Path [Path 1-1]:} \\ &= (\text{SONW} + \text{SNW}) = (66642 + 804795) = 871437 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 2-2]:} \\ &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (66642 + 0 + 504306 + 0) = 570948 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 3-3]:} \\ &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (66642 + 504306 + 0) = 570948 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

$$\begin{aligned} \text{Path 1-1} &= 871437 \text{ Kgf, must exceed } W = 412675 \text{ Kgf} & \text{or } W1 &= 210757 \text{ Kgf} \\ \text{Path 2-2} &= 570948 \text{ Kgf, must exceed } W = 412675 \text{ Kgf} & \text{or } W2 &= 561919 \text{ Kgf} \\ \text{Path 3-3} &= 570948 \text{ Kgf, must exceed } W = 412675 \text{ Kgf} & \text{or } W3 &= 561919 \text{ Kgf} \end{aligned}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 16.7246 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 2549.0000 mm

PV Elite 28 Licensee: SL User
FileName : 106-D
Nozzle Calcs.: U1_600

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Input, Nozzle Desc: K575

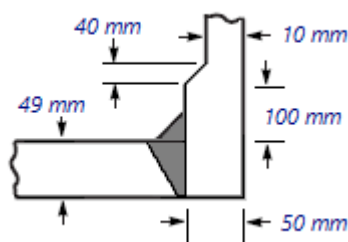
From: 40

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	49.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.04	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		75.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K575

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	75.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
= 45.6284 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4621.0 \cdot 40.5 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$
= 1.5184 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6893 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $40.0 / 0.5773$
= 69.2821 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	203.0364	mm
Parallel to Vessel Wall	Rn+tn+t	101.5182	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm

Weld Strength Reduction Factor [fr1]:

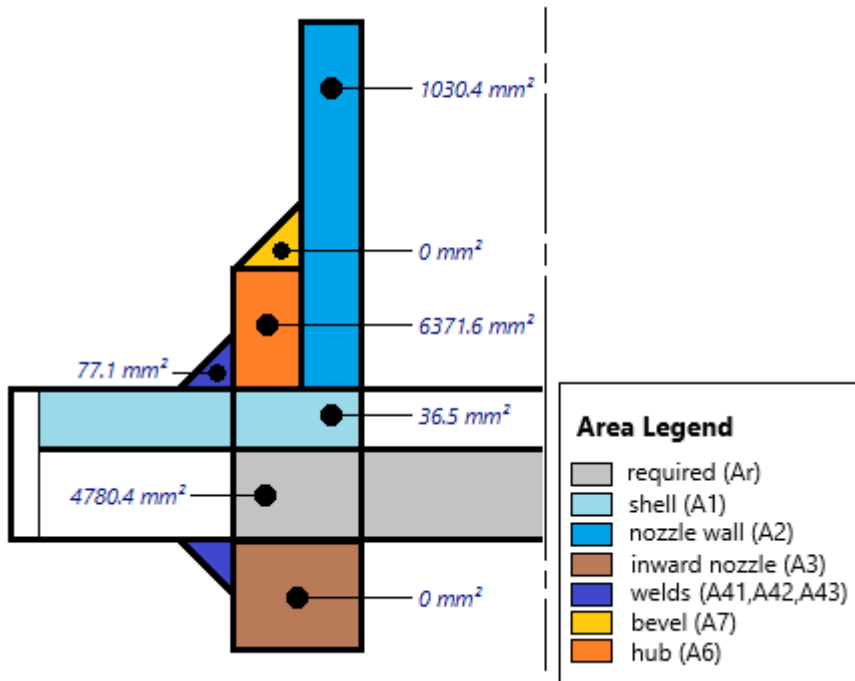
= min(1, S_n / S_v)
= min(1, $126.0 / 137.3$)
= 0.918

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)
= min(1, $126.0 / 137.3$)
= 0.918

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(0.918, 1.0)
= 0.918



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	47.793	4.773	NA
Area in Shell	A1	0.365	36.256	NA
Area in Nozzle Wall	A2	10.308	11.867	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.771	0.771	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	63.733	63.733	NA
TOTAL AREA AVAILABLE	Atot	75.178	112.627	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.59 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (97.0364 \cdot 45.6284 \cdot 1.0 + 2 \cdot 47.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 47.793 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 106.0(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 47.0$
 $(1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918)$
 $= 0.365 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 86.78)(7.0 - 1.52) \cdot 0.918/\sin(57.9)$
 $= 10.308 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can/0.707)^2 - \text{Area Lost}) \cdot fr2$
 $= (10.0^2 - 0.16) \cdot 0.918 + (0.0^2 - 0.0) \cdot 0.918$

$$= 0.771 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(86.8, 500.0, 100.0)) \cdot (50.0 - 10.0) \cdot 0.918 \\ &= 63.733 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 10.0, tr = 1.384, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.198, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -16 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4621.00/10340.00 = 0.447

Weld Size Calculations, Description: K575

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$
$$= \max(0, (47.7929 - 0.3652 + 2 \cdot 47.0 \cdot 0.918 \cdot (1.0 \cdot 46.0 - 45.6284)) \cdot 137)$$
$$= 66845.93 \text{ Kg}$$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
$$= (A_2 + A_4 - (W_i - C_{an}/.707)^2 \cdot f_{r2}) \cdot S_v$$
$$= (74.0413 + 0.7711 - 0.0 \cdot 0.92) \cdot 137$$
$$= 104734.55 \text{ Kg}$$

Weld Load [W2]:
$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$
$$= (74.0413 + 0.0 + 0.7711 + (39.6946)) \cdot 137$$
$$= 160305.55 \text{ Kg}$$

Weld Load [W3]:
$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$
$$= (74.0413 + 0.0 + 0.7711 + 0.0 + (39.6946)) \cdot 137$$
$$= 160305.55 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 209.6465 \cdot 10.0 \cdot 0.49 \cdot 126 \\ &= 20738. \text{ Kg} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 76.6707) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 101845. \text{ Kg} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 209.6465 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 156933. \text{ Kg} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (20738 + 101845) = 122583 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (20738 + 0 + 156933 + 0) = 177671 \text{ Kg} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (20738 + 156933 + 0) = 177671 \text{ Kg} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 122582 Kg, must exceed W = 66845 Kg	or W1 = 104734 Kg
Path 2-2 = 177670 Kg, must exceed W = 66845 Kg	or W2 = 160305 Kg
Path 3-3 = 177670 Kg, must exceed W = 66845 Kg	or W3 = 160305 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K575

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	75.000 mm.
Actual Thickness Used in Calculation	10.000 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0) \\ &= 45.6284 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 4621.0 \cdot 40.5 / (126 \cdot 1.0 - 0.6 \cdot 4621.0) \\ &= 1.5184 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.6893 mm



Intermediate Hub Nozzle Calculations:

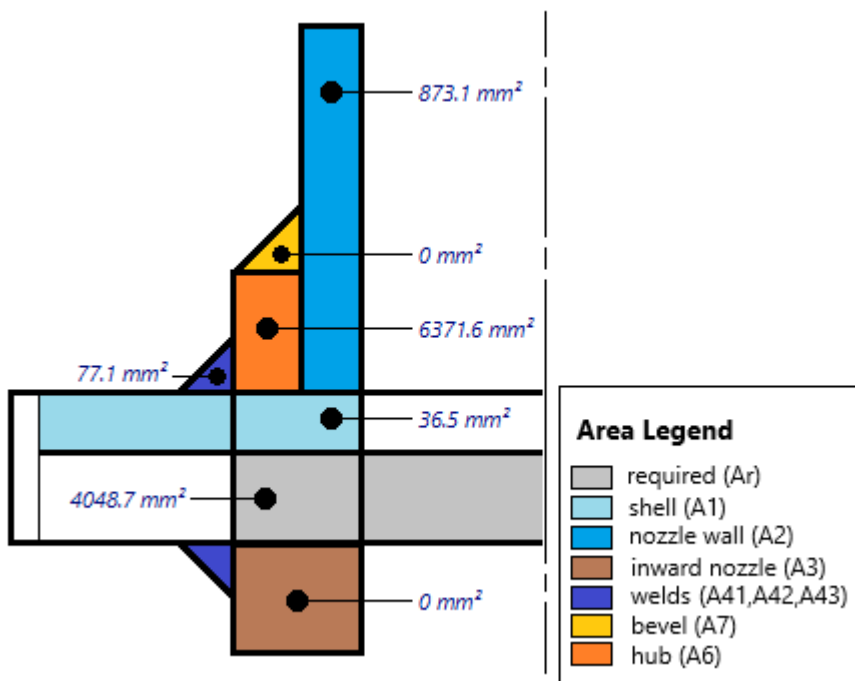
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.0000 mm
 Parallel to Vessel Wall Rn+tn+t 93.5000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 86.7821 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	40.476	4.043	NA
Area in Shell	A1	0.365	36.256	NA
Area in Nozzle Wall	A2	8.734	10.055	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.771	0.771	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	63.733	63.733	NA
TOTAL AREA AVAILABLE	Atot	73.604	110.815	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (81.0 \cdot 45.6284 \cdot 1.0 + 2 \cdot 47.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 40.476 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 106.0(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 47.0 \\ &\quad (1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918) \\ &= 0.365 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot tnp)(tn - trn)fr2 \\ &= (2 \cdot 86.78)(7.0 - 1.52) \cdot 0.918 \\ &= 8.734 \text{ cm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= (Wo^2 - \text{Area Lost}) \cdot fr2 + ((Wi - can/0.707)^2 - \text{Area Lost}) \cdot fr2 \\ &= (10.0^2 - 0.16) \cdot 0.918 + (0.0^2 - 0.0) \cdot 0.918 \\ &= 0.771 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(86.8, 500.0, 100.0)) \cdot (50.0 - 10.0) \cdot 0.918 \\ &= 63.733 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5184 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 48.6284 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 48.6284 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0178 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.02, \max(48.6284, 4.5)] \\ &= 8.0200 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.5184, 8.02) \\ &= 8.0200 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: K575

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (40.4758 - 0.3652 + 2 \cdot 47.0 \cdot 0.918 \cdot \\ &\quad (1.0 \cdot 46.0 - 45.6284)) \cdot 137) \\ &= 56602.18 \text{ Kg} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (Wi - Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (72.4672 + 0.7711 - 0.0 \cdot 0.92) \cdot 137 \\ &= 102530.98 \text{ Kg} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (72.4672 + 0.0 + 0.7711 + (39.6946)) \cdot 137 \end{aligned}$$



= 158101.98 Kgf

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\ &= (72.4672 + 0.0 + 0.7711 + 0.0 + (39.6946)) \cdot 137 \\ &= 158101.98 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 175.0 \cdot 10.0 \cdot 0.49 \cdot 126 \\ &= 17311. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 64.0) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 85014. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 175.0 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 130998. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (17311 + 85014) = 102324 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (17311 + 0 + 130998 + 0) = 148309 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (17311 + 130998 + 0) = 148309 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 102324 Kgf, must exceed W = 56602 Kgf	or W1 = 102530 Kgf
Path 2-2 = 148308 Kgf, must exceed W = 56602 Kgf	or W2 = 158101 Kgf
Path 3-3 = 148308 Kgf, must exceed W = 56602 Kgf	or W3 = 158101 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 61.0310 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 619.7386 mm

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Input, Nozzle Desc: K450

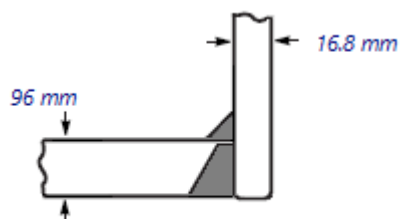
From: 50

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5350.00	mm
Design Length of Section	L	3649.3335	mm
Shell Finished (Minimum) Thickness	t	96.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		900.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F11	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		96.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K450

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.800 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4621.0 \cdot 2678.0 / (137 \cdot 1.0 - 0.6 \cdot 4621.0)$
= 91.9999 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4621.0 \cdot 28.4 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$
= 1.0651 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5520 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	270.4000 mm
Parallel to Vessel Wall	Rn+tn+t	135.2000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K450.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.0651 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 94.9999 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 94.9999 mm
Wall Thickness, shell/head, external pressure	trb2	= 5.0183 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.0183 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(95.0, 5.0183)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.0651, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4621.00/10340.00 = 0.447$

Weld Size Calculations, Description: K450

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3329 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 446.3329 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 135.20 mm

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Input, Nozzle Desc: M1600

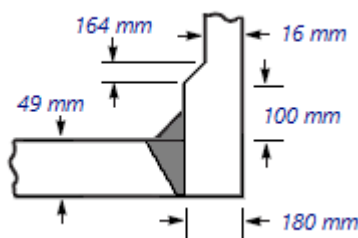
From: 60

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	49.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.04	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	164.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.





Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: M1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	600.000	mm.
Actual Thickness Used in Calculation	16.000	mm

Note:
Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
= 45.6284 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4621.0 \cdot 300.0 / (126 \cdot 1.0 + 0.4 \cdot 4621.0)$
= 10.8410 mm

Required Nozzle thickness under External Pressure per UG-28 : 2.6615 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $164.0 / 0.5773$
= 284.0565 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

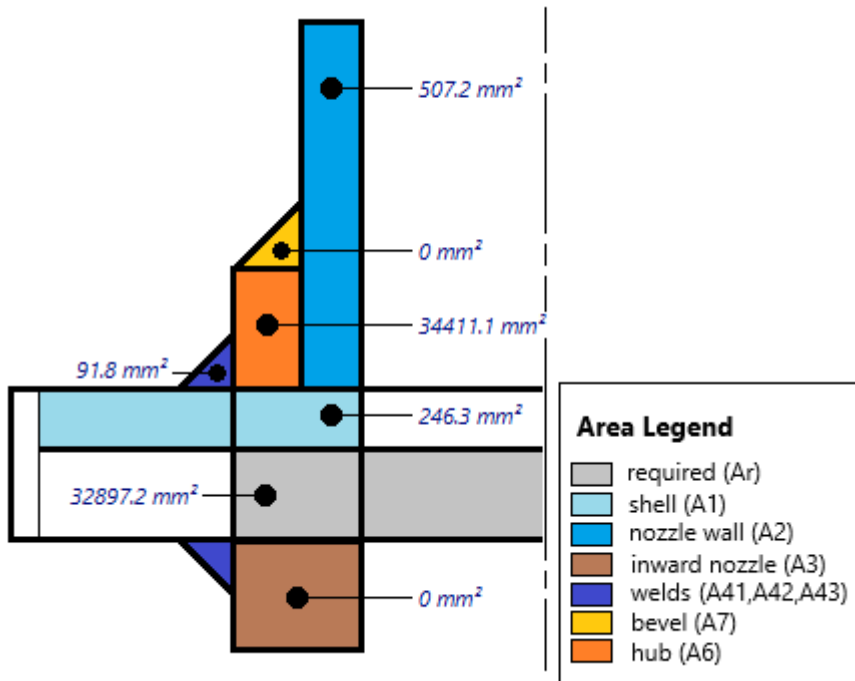
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1383.7292	mm
Parallel to Vessel Wall, opening length	d	691.8646	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	115.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $126.0 / 137.3$)
= 0.918

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $126.0 / 137.3$)
= 0.918

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.918, 1.0)
= 0.918



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	328.931	32.853	NA
Area in Shell	A1	2.463	244.491	NA
Area in Nozzle Wall	A2	5.080	24.327	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		0.918	0.918	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	344.207	344.207	NA
TOTAL AREA AVAILABLE	Atot	352.668	613.943	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.06 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (691.8646 \cdot 45.6284 \cdot 1.0 + 2 \cdot 177.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$$

$$= 328.931 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 691.865(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 177.0$$

$$(1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918)$$

$$= 2.463 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$$

$$= (2 \cdot 115.0)(13.0 - 10.84)0.918/\sin(63.8)$$

$$= 5.080 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.918 + (0.0)^2 \cdot 0.918$$



= 0.918 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(115.0, 1000.0, 100.0)) \cdot (180.0 - 16.0) \cdot 0.918 \\ &= 344.207 \text{ cm}^2 \text{ Includes Bevel Area of } 43.101 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 16.0, tr = 9.914, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.763, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -16 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4621.00/10340.00 = 0.447

Weld Size Calculations, Description: M1600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (328.9309 - 2.4629 + 2 * 177.0 * 0.918 *
(1.0 * 46.0 - 45.6284))137)
= 458733.62 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²*fr2)*Sv
= (349.2872 + 0.918 - 0.0 * 0.92) * 137
= 490274.41 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (349.2872 + 0.0 + 0.918 + (149.4882)) * 137
= 699552.38 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (349.2872 + 0.0 + 0.918 + 0.0 + (149.4882)) * 137
= 699552.38 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 723.2034 \cdot 10.0 \cdot 0.49 \cdot 126 \\ &= 71538. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 353.767) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 1769706. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 723.2034 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 541360. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (71538 + 1769706) = 1841244 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (71538 + 0 + 541361 + 0) = 612899 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (71538 + 541361 + 0) = 612899 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1841244 Kgf, must exceed W = 458733 Kgf or W1 = 490274 Kgf
Path 2-2 = 612898 Kgf, must exceed W = 458733 Kgf or W2 = 699552 Kgf
Path 3-3 = 612898 Kgf, must exceed W = 458733 Kgf or W3 = 699552 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: M1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	600.000 mm.
Actual Thickness Used in Calculation	16.000 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0) \\ &= 45.6284 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\ &= 4621.0 \cdot 300.0 / (126 \cdot 1.0 + 0.4 \cdot 4621.0) \\ &= 10.8410 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 2.6615 mm



Intermediate Hub Nozzle Calculations:

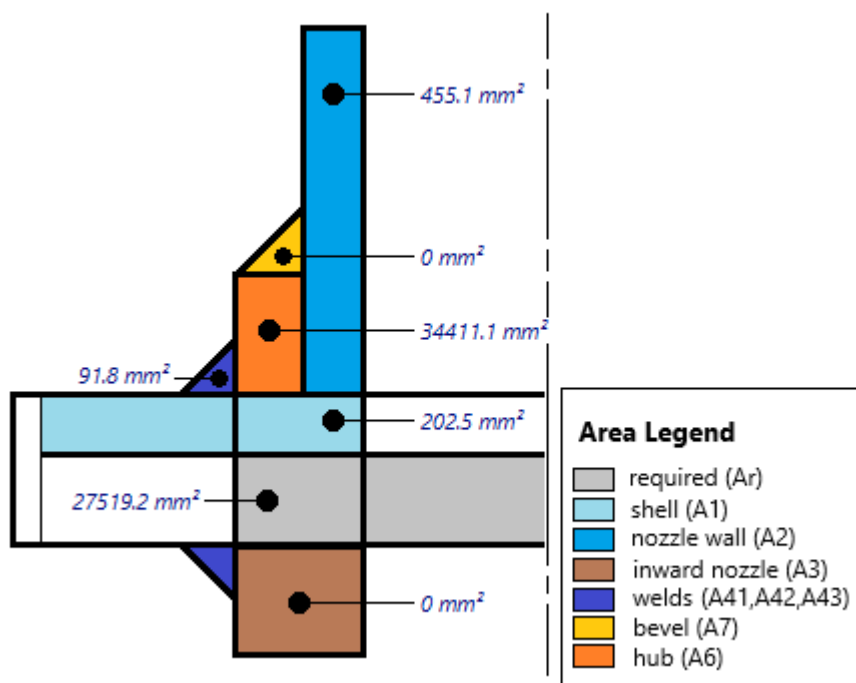
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 164.0/0.5773$
 $= 284.0565 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1148.0001 mm
 Parallel to Vessel Wall, opening length d 574.0001 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 115.0000 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	275.151	27.482	NA
Area in Shell	A1	2.025	201.016	NA
Area in Nozzle Wall	A2	4.559	21.829	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.918	0.918	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	344.207	344.207	NA
TOTAL AREA AVAILABLE	Atot	351.708	567.970	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (574.0 \cdot 45.6284 \cdot 1.0 + 2 \cdot 177.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 275.151 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 574.0(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 177.0 \\ &\quad (1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918) \\ &= 2.025 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot tlnp)(tn - trn)fr2 \\ &= (2 \cdot 115.0)(13.0 - 10.84)0.918 \\ &= 4.559 \text{ cm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2 \\ &= 10.0^2 \cdot 0.918 + (0.0)^2 \cdot 0.918 \\ &= 0.918 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(115.0, 1000.0, 100.0)) \cdot (180.0 - 16.0) \cdot 0.918 \\ &= 344.207 \text{ cm}^2 \text{ Includes Bevel Area of } 43.101 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

$$\text{Wall Thickness for Internal/External pressures} \quad ta = 13.8410 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= ta \\ &= 13.8410 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Weld Size Calculations, Description: M1600

$$\text{Intermediate Calc. for nozzle/shell welds} \quad Tmin \quad 19.0000 \text{ mm}$$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (275.1512 - 2.0249 + 2 \cdot 177.0 \cdot 0.918 \cdot \\ &\quad (1.0 \cdot 46.0 - 45.6284)) \cdot 137) \\ &= 384057.06 \text{ Kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (348.7655 + 0.918 - 0.0 \cdot 0.92) \cdot 137 \\ &= 489544.00 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (348.7655 + 0.0 + 0.918 + (149.4882)) \cdot 137 \\ &= 698822.00 \text{ Kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (348.7655 + 0.0 + 0.918 + 0.0 + (149.4882)) \cdot 137 \\ &= 698822.00 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 600.0 \cdot 10.0 \cdot 0.49 \cdot 126 \end{aligned}$$



= 59351. Kgf

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 293.5) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 1468222. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 600.0 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 449135. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (59351 + 1468222) = 1527574 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (59351 + 0 + 449135 + 0) = 508487 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (59351 + 449135 + 0) = 508487 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1527573 Kgf, must exceed W = 384057 Kgf	or W1 = 489544 Kgf
Path 2-2 = 508486 Kgf, must exceed W = 384057 Kgf	or W2 = 698822 Kgf
Path 3-3 = 508486 Kgf, must exceed W = 384057 Kgf	or W3 = 698822 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 232.6724 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1291.3801 mm

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Input, Nozzle Desc: K350

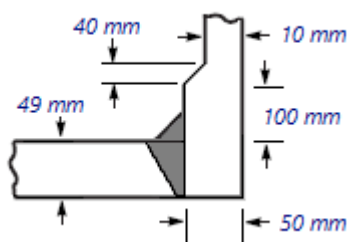
From: 60

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	49.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.04	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	2000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
= 45.6284 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4621.0 \cdot 28.0 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$
= 1.0498 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6836 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 40.0/0.5773
= 69.2821 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

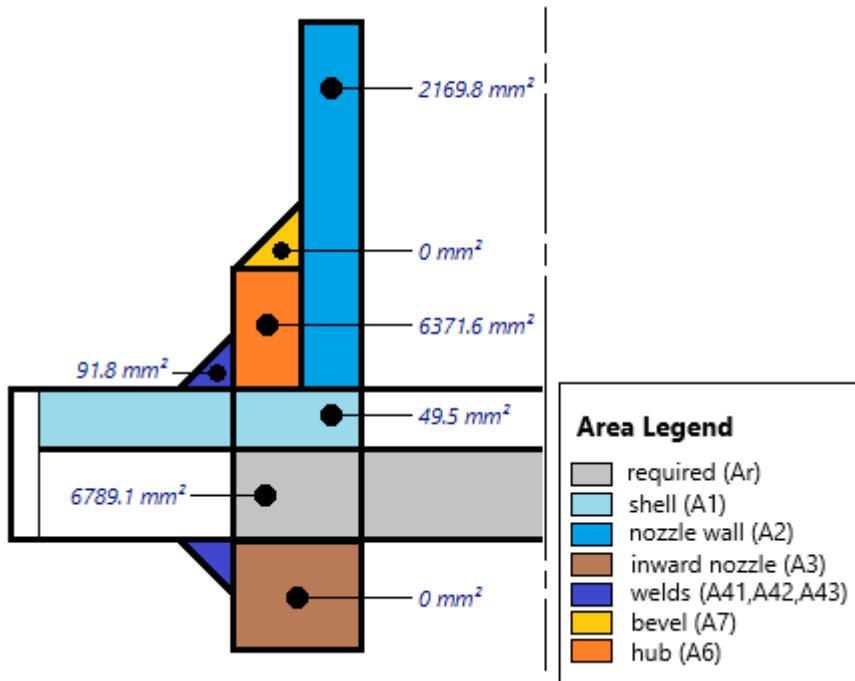
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	282.1195	mm
Parallel to Vessel Wall, opening length	d	141.0598	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 126.0/137.3)
= 0.918

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 126.0/137.3)
= 0.918

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.918, 1.0)
= 0.918



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	67.880	6.780	NA
Area in Shell	A1	0.495	49.188	NA
Area in Nozzle Wall	A2	21.705	23.041	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.918	0.918	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	63.733	63.733	NA
TOTAL AREA AVAILABLE	Atot	86.852	136.880	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 23.39 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (141.0598 \cdot 45.6284 \cdot 1.0 + 2 \cdot 47.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 67.880 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 141.06(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 47.0$
 $(1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918)$
 $= 0.495 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 86.78)(7.0 - 1.05)0.918/\sin(25.9)$
 $= 21.705 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.918 + (0.0)^2 \cdot 0.918$



= 0.918 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(86.8, 2000.0, 100.0)) \cdot (50.0 - 10.0) \cdot 0.918 \\ &= 63.733 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 10.0, tr = 0.957, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.137, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -16 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4621.00/10340.00 = 0.447

Weld Size Calculations, Description: K350

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (67.8801 - 0.4955 + 2 • 47.0 • 0.918 •
(1.0 • 46.0 - 45.6284)) 137)
= 94784.85 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (85.4386 + 0.918 - 0.0 • 0.92) • 137
= 120896.06 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (85.4386 + 0.0 + 0.918 + (39.6946)) • 137
= 176467.05 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (85.4386 + 0.0 + 0.918 + 0.0 + (39.6946)) • 137
= 176467.05 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 377.8387 \cdot 10.0 \cdot 0.49 \cdot 126 \\ &= 37375. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 129.7246) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 172318. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 377.8387 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 282835. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (37375 + 172318) = 209693 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (37375 + 0 + 282835 + 0) = 320210 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (37375 + 282835 + 0) = 320210 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 209693 Kgf, must exceed W = 94784 Kgf	or W1 = 120896 Kgf
Path 2-2 = 320209 Kgf, must exceed W = 94784 Kgf	or W2 = 176467 Kgf
Path 3-3 = 320209 Kgf, must exceed W = 94784 Kgf	or W3 = 176467 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.000 mm.
Actual Thickness Used in Calculation	10.000 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0) \\ &= 45.6284 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 4621.0 \cdot 28.0 / (126 \cdot 1.0 - 0.6 \cdot 4621.0) \\ &= 1.0498 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.6836 mm



Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 40.0/0.5773
= 69.2821 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	162.0000	mm
Parallel to Vessel Wall	Rn+tn+t	81.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K350.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.0498	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 48.6284	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 48.6284	mm
Wall Thickness, shell/head, external pressure	trb2	= 4.0178	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.5200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.52, max(48.6284, 4.5)]
= 7.5200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.0498, 7.52)
= 7.5200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: K350

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 208.4659 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 2332.1472 mm

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Input, Nozzle Desc: K150

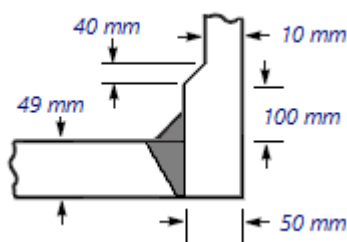
From: 60

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	49.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.04	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	5000.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
= 45.6284 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4621.0 \cdot 28.0 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$
= 1.0498 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6749 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 40.0/0.5773
= 69.2821 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

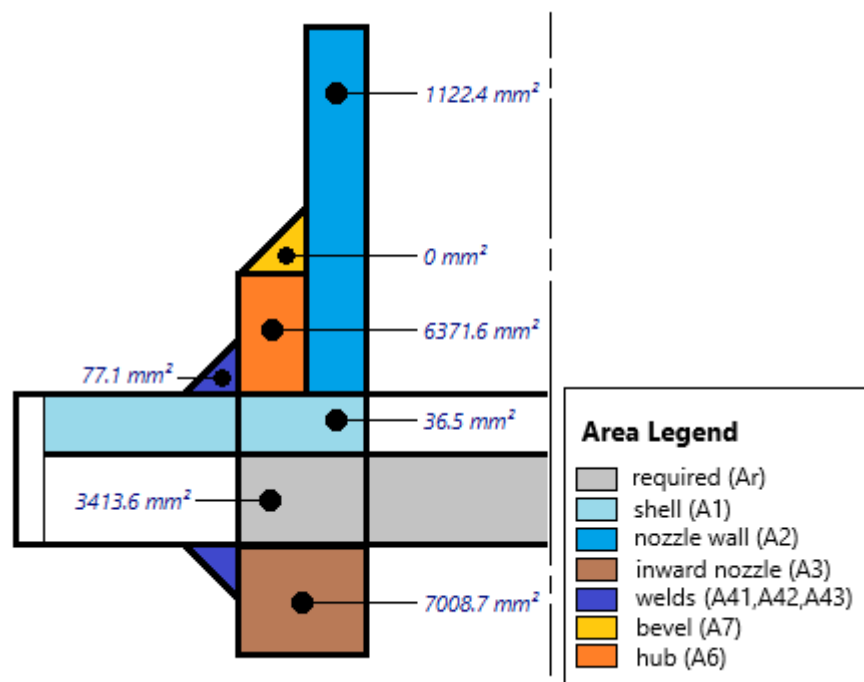
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	173.0827	mm
Parallel to Vessel Wall	Rn+tn+t	86.5413	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm
Normal to Vessel Wall, Inward		10.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 126.0/137.3)
= 0.918

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 126.0/137.3)
= 0.918

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.918, 1.0)
= 0.918



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	34.126	3.408	NA
Area in Shell	A1	0.365	36.256	NA
Area in Nozzle Wall	A2	11.227	11.935	NA
Area in Inward Nozzle	A3	70.107	70.107	NA
Area in Welds	A41+A42+A43	0.771	0.771	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	63.733	63.733	NA
TOTAL AREA AVAILABLE	Atot	146.203	182.801	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.59 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (67.0826 \cdot 45.6284 \cdot 1.0 + 2 \cdot 47.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 34.126 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 106.0(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 47.0$
 $(1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918)$
 $= 0.365 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 86.78)(7.0 - 1.05)0.918/\sin(57.6)$
 $= 11.227 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 44.0 \cdot (86.7821) \cdot 0.918$



$$= 70.107 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= (W_o^2 - \text{Area Lost}) \cdot fr_2 + ((W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr_2 \\ &= (10.0^2 - 0.16) \cdot 0.918 + (0.0^2 - 0.0) \cdot 0.918 \\ &= 0.771 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(86.8, 1000.0, 100.0)) \cdot (50.0 - 10.0) \cdot 0.918 \\ &= 63.733 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 0.957$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.137$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -16°C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18°C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48°C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4621.00/10340.00 = 0.447$

Weld Size Calculations, Description: K150

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (34.1255 - 0.3652 + 2 \cdot 47.0 \cdot 0.918 \cdot (1.0 \cdot 46.0 - 45.6284)) \cdot 137) \\ &= 47712.04 \text{ Kg} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (W_i - \text{Can}/0.707)^2 \cdot fr_2) \cdot S_v \\ &= (74.9605 + 0.7711 - 0.0 \cdot 0.92) \cdot 137 \\ &= 106021.52 \text{ Kg} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (74.9605 + 70.1065 + 0.7711 + (39.6946)) \cdot 137 \\ &= 259739.09 \text{ Kg} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (74.9605 + 70.1065 + 0.7711 + 0.0 + (39.6946)) \cdot 137 \end{aligned}$$



= 259739.09 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 179.6857 \cdot 10.0 \cdot 0.49 \cdot 126 \\ &= 17774. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 61.6921) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 81948. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 179.6857 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 134505. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (17774 + 81948) = 99722 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (17774 + 0 + 134505 + 0) = 152280 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (17774 + 134505 + 0) = 152280 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 99722 Kgf, must exceed W = 47712 Kgf	or W1 = 106021 Kgf
Path 2-2 = 152279 Kgf, must exceed W = 47712 Kgf	or W2 = 259739 Kgf
Path 3-3 = 152279 Kgf, must exceed W = 47712 Kgf	or W3 = 259739 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.000 mm.
Actual Thickness Used in Calculation	10.000 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0) \\ &= 45.6284 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$



$$= 4621.0 \cdot 28.0 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$$
$$= 1.0498 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.6749 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$
$$= 40.0 / 0.5773$$
$$= 69.2821 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	162.0000	mm
Parallel to Vessel Wall	Rn+tn+t	81.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm
Normal to Vessel Wall, Inward		10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K150.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.0498	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 48.6284	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 48.6284	mm
Wall Thickness, shell/head, external pressure	trb2	= 4.0178	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.5200	mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[7.52, \max(48.6284, 4.5)]$$
$$= 7.5200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(4.0498, 7.52)$$
$$= 7.5200 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: K150

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 51.9943 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 6058.7080 mm

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Input, Nozzle Desc: N1750

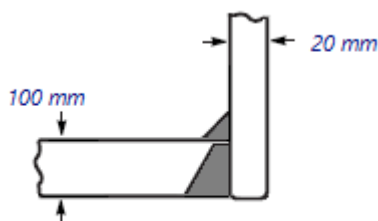
From: 60

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	100.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		750.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1750

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 750.000 mm.
Actual Thickness Used in Calculation 20.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
= 45.6284 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4621.0 \cdot 378.0 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$
= 14.1758 mm

Required Nozzle thickness under External Pressure per UG-28 : 3.1376 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1512.0000	mm
Parallel to Vessel Wall, opening length	d	756.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	42.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $126.0 / 137.3$)
= 0.918

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $126.0 / 137.3$)
= 0.918

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.918, 1.0)
= 0.918

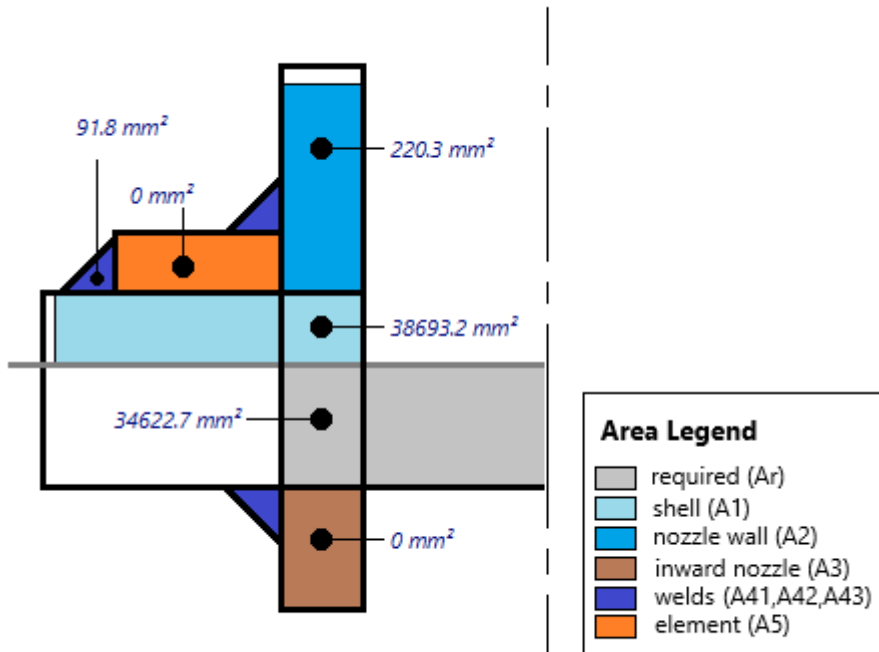


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	346.227	34.580	NA
Area in Shell	A1	386.932	661.956	NA
Area in Nozzle Wall	A2	2.203	10.814	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.918	0.918	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	390.053	673.688	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (756.0 \cdot 45.6284 \cdot 1.0 + 2 \cdot 17.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 346.227 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 756.0(1.0 \cdot 97.0 - 1.0 \cdot 45.6284) - 2 \cdot 17.0$
 $(1.0 \cdot 97.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918)$
 $= 386.932 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 42.5)(17.0 - 14.18) \cdot 0.9178$
 $= 2.203 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9178 + (0.0)^2 \cdot 0.9178$
 $= 0.918 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 17.1758$ mm
Wall Thickness per UG16(b), $t_{r16b} = 4.5000$ mm
Wall Thickness, shell/head, internal pressure $t_{rb1} = 48.6284$ mm
Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 48.6284$ mm
Wall Thickness, shell/head, external pressure $t_{rb2} = 4.0178$ mm
Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.5000$ mm
Wall Thickness per table UG-45 $t_{b3} = 11.3312$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$
 $= \min[11.331, \max(48.6284, 4.5)]$
 $= 11.3312$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(17.1758, 11.3312)$
 $= 17.1758$ mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	69.0,	Allowable	:	126.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	261.0 N./mm ²	Passed
Occasional	:	50.2,	Allowable	:	167.6 N./mm ²	Passed
Shear	:	11.9,	Allowable	:	88.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, Curve: B**

Govrn. thk, $t_g = 20.0$, $t_r = 12.918$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio $= t_r * E^* / (t_g - c) = 0.76$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-21 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -21 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4621.00/10340.00 = 0.447$

Weld Size Calculations, Description: N1750

Intermediate Calc. for nozzle/shell Welds $T_{min} = 17.0000$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r))S_v)$



$$= \max(0, (346.227 - 386.9325 + 2 \cdot 17.0 \cdot 0.9178 \cdot (1.0 \cdot 97.0 - 45.6284)) \cdot 137)$$

$$= \max(0, -34545.12) \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr2}) \cdot S_v$$

$$= (2.2032 + 0.0 + 0.9178 - 0.0 \cdot 0.92) \cdot 137$$

$$= 4369.16 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S_v$$

$$= (2.2032 + 0.0 + 0.9178 + (30.2675)) \cdot 137$$

$$= 46742.48 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S$$

$$= (2.2032 + 0.0 + 0.9178 + 0.0 + (30.2675)) \cdot 137$$

$$= 46742.48 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 790.0 \cdot 10.0 \cdot 0.49 \cdot 126$$

$$= 78124. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 386.5) \cdot (20.0 - 3.0) \cdot 0.7 \cdot 126$$

$$= 185647. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 790.0 \cdot (49.0 - 0.0) \cdot 0.74 \cdot 137$$

$$= 629929. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (78124 + 185647) = 263771 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (78124 + 0 + 629929 + 0) = 708053 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (78124 + 629929 + 0) = 708053 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 263771 Kgf, must exceed W = 0 Kgf	or W1 = 4369 Kgf
Path 2-2 = 708052 Kgf, must exceed W = 0 Kgf	or W2 = 46742 Kgf
Path 3-3 = 708052 Kgf, must exceed W = 0 Kgf	or W3 = 46742 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 29.0606 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 1129.0607 mm

Input Echo, WRC107/537 Item 1, Description: N1750 :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5398.000	mm
Vessel Thickness	Tv	100.000	mm
Design Temperature	T1	454.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	137.29	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	750.000	mm
Nozzle Thickness	Tn	20.000	mm
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	126.00	N./mm ²
Design Internal Pressure	Dp	4621.000	KPa.
Include Pressure Thrust		No	

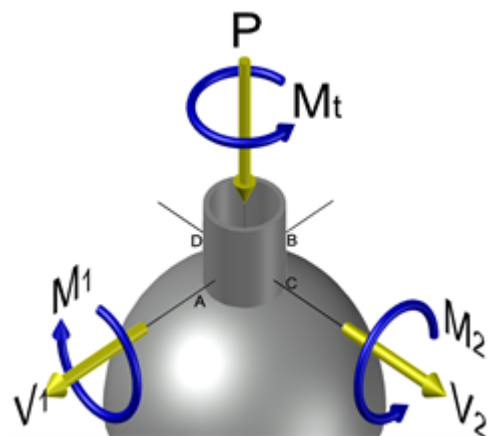
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-6880.0	Kgf
Longitudinal Shear (SUS)	(V1) V1	4970.0	Kgf
Circumferential Shear (SUS)	(Vc) V2	4970.0	Kgf
Circumferential Moment (SUS)	(Mc) M1	11468.0	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	11468.0	Kg-m.
Torsional Moment (SUS)	Mt	13188.1	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 790.0 + 2 \cdot 1.65 \cdot \sqrt{(2750.5 (100.0 - 3.0))}$$

$$= 2494.534 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-6880.0	Kgf
Circumferential Shear	(VC) V2	4970.0	Kgf
Longitudinal Shear	(VL) V1	4970.0	Kgf
Circumferential Moment	(MC) M1	11468.0	Kg-m.
Longitudinal Moment	(ML) M2	11468.0	Kg-m.
Torsional Moment	MT	13188.1	Kg-m.

Dimensionless Parameters: U = 0.76 TAU = 22.74 RHO = 5.71

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 7	0.02888	(A,B,C,D)
M(x) / P	SP 7	0.00979	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 7	0.06548	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 7	0.02145	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 7	0.06548	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 7	0.02145	(A,B,C,D)
N(y) * T / P	SP 7	0.24978	(A,B,C,D)
M(y) / P	SP 7	0.04178	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.33432	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.11021	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.33432	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.11021	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad. Bend. P		0.4	-0.4	0.4	-0.4	0.4	-0.4	0.4	-0.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-3.0	3.0	3.0	-3.0
Rad. Memb. ML		-1.5	-1.5	1.5	1.5	0.0	0.0	0.0	0.0
Rad. Bend. ML		-3.0	3.0	3.0	-3.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-3.9	1.2	5.1	-1.7	-3.9	1.2	5.1	-1.7
Tang. Memb. P		1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Tang. Bend. P		1.8	-1.8	1.8	-1.8	1.8	-1.8	1.8	-1.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-7.7	-7.7	7.7	7.7
Tang. Bend. MC		0.0	0.0	0.0	0.0	-15.3	15.3	15.3	-15.3
Tang. Memb. ML		-7.7	-7.7	7.7	7.7	0.0	0.0	0.0	0.0
Tang. Bend. ML		-15.3	15.3	15.3	-15.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-19.4	7.6	26.6	-7.6	-19.4	7.6	26.6	-7.6
Shear VC		0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT		1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Tot. Shear		1.8	1.8	1.0	1.0	1.0	1.0	1.8	1.8
Str. Int.		19.6	8.0	26.7	7.7	19.5	7.7	26.8	8.1



WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	64.4	64.4	64.4	64.4	64.4	64.4	64.4	64.4
Rad.	Pl (SUS)	-1.3	-1.3	1.7	1.7	-1.3	-1.3	1.7	1.7
Rad.	Q (SUS)	-2.6	2.6	3.4	-3.4	-2.6	2.6	3.4	-3.4
Long.	Pm (SUS)	64.4	64.4	64.4	64.4	64.4	64.4	64.4	64.4
Long.	Pl (SUS)	-5.9	-5.9	9.5	9.5	-5.9	-5.9	9.5	9.5
Long.	Q (SUS)	-13.5	13.5	17.1	-17.1	-13.5	13.5	17.1	-17.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear	Q (SUS)	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Pm (SUS)		64.4	64.4	64.4	64.4	64.4	64.4	64.4	64.4
Pm+Pl (SUS)		63.1	63.1	74.0	74.0	63.1	63.1	74.0	74.0
Pm+Pl+Q (Total)		60.8	72.4	91.1	62.9	60.6	72.1	91.2	63.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	64.42	137.29	Passed
Pm+Pl (SUS)	73.97	205.94	Passed
Pm+Pl+Q (TOTAL)	91.19	411.88	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K250

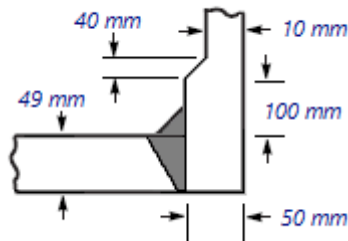
From: 60

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	49.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-182 F11	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	5000.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	10.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
 $= 45.6284$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4621.0 \cdot 28.4 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$
 $= 1.0651$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4679 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

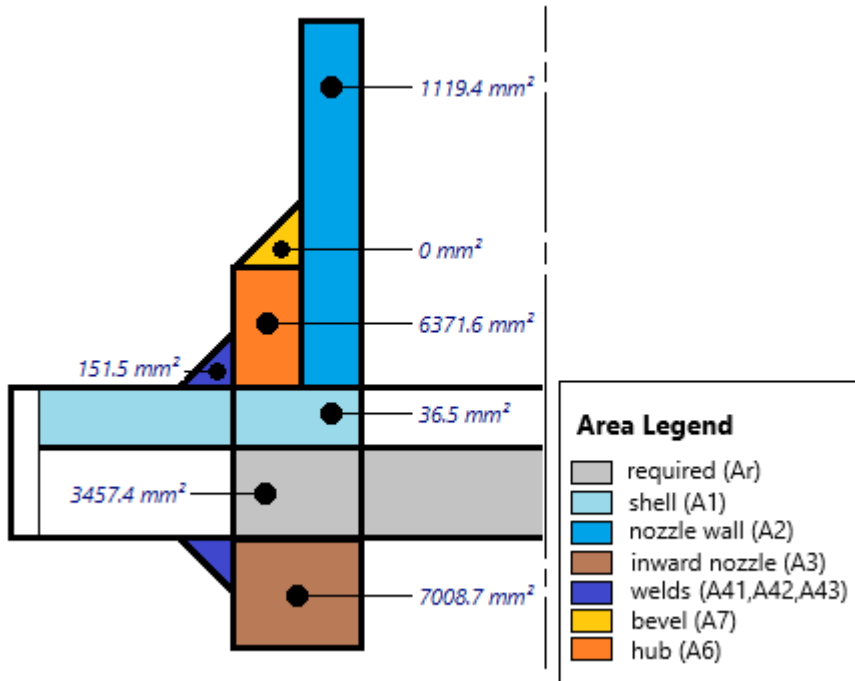
Parallel to Vessel Wall (Diameter Limit)	D1	174.0414	mm
Parallel to Vessel Wall	Rn+tn+t	87.0207	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm
Normal to Vessel Wall, Inward		10.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 126.0 / 137.3)$
 $= 0.918$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 126.0 / 137.3)$
 $= 0.918$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$

= min(0.918, 1.0)
 = 0.918



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	34.574	3.453	NA
Area in Shell	A1	0.365	36.247	NA
Area in Nozzle Wall	A2	11.194	12.320	NA
Area in Inward Nozzle	A3	70.087	70.087	NA
Area in Welds	A41+A42+A43	1.515	1.515	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	63.716	63.716	NA
TOTAL AREA AVAILABLE	Atot	146.877	183.885	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.59 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (68.0414 \cdot 45.6284 \cdot 1.0 + 2 \cdot 47.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 34.574 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 106.0 (1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 47.0$
 $(1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918)$
 $= 0.365 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp) (tn - trn) fr2 / \sin(\alpha3)$
 $= (2 \cdot 86.78) (7.0 - 1.07) 0.9178 / \sin(57.6)$
 $= 11.194 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.



Area Available in Inward Nozzle [A3]:

$$\begin{aligned} &= 2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2 \\ &= 2 \cdot 44.0 \cdot (86.7821) \cdot 0.9178 \\ &= 70.087 \text{ cm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi\text{-}can/0.707)^2 - \text{Area Lost}) \cdot fr2 \\ &= (10.0^2 - 0.16) \cdot 0.9178 + (9.7573^2 - 0.1412) \cdot 0.9178 \\ &= 1.515 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(86.8, 300.0, 100.0)) \cdot (50.0 - 10.0) \cdot 0.918 \\ &= 63.716 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 10.0, tr = 0.971, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.139, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -16 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4621.00/10340.00 = 0.447

Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (34.5739 - 0.3651 + 2 \cdot 47.0 \cdot 0.9178 \cdot (1.0 \cdot 46.0 - 45.6284)) \cdot 137) \\ &= 48339.75 \text{ Kg} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi\text{-}Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (74.9096 + 1.5151 - 95.2053 \cdot 0.92) \cdot 137 \\ &= 105768.57 \text{ Kg} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (74.9096 + 70.0871 + 1.5151 + (39.6836)) \cdot 137 \\ &= 260666.67 \text{ Kg} \end{aligned}$$



Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (74.9096 + 70.0871 + 1.5151 + 0.0 + (39.6836)) * 137
= 260666.67 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 180.6451 * 10.0 * 0.49 * 126
= 17864. Kgf

Shear, Inward Nozzle Weld [Sinw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 180.6451 * 9.7573 * 0.49 * 126
= 17431. Kgf

Shear, Nozzle Wall [Snw]:
= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn
= (3.1416 * 62.1716) * (50.0 - 3.0) * 0.7 * 126
= 82562. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) * Dlo * (T - cas) * 0.74 * Sng
= (3.1416/2.0) * 180.6451 * (49.0 - 3.0) * 0.74 * 137
= 135223. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (17864 + 82562) = 100426 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (17864 + 0 + 135223 + 17431) = 170518 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (17864 + 135223 + 17431) = 170518 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 100426 Kgf, must exceed W = 48339 Kgf	or W1 = 105768 Kgf
Path 2-2 = 170518 Kgf, must exceed W = 48339 Kgf	or W2 = 260666 Kgf
Path 3-3 = 170518 Kgf, must exceed W = 48339 Kgf	or W3 = 260666 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	10.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= P*R/(2*Sv*E-0.2*P) per UG-27 (d)
= 4621.0*2702.0/(2*137*1.0-0.2*4621.0)
= 45.6284 mm



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4621.0 \cdot 28.4 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$
 $= 1.0651 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4679 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	162.8000	mm
Parallel to Vessel Wall	Rn+tn+t	81.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm
Normal to Vessel Wall, Inward		10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K250.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.0651	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 48.6284	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 48.6284	mm
Wall Thickness, shell/head, external pressure	trb2	= 4.0178	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.5200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.52, \max(48.6284, 4.5)]$
 $= 7.5200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.0651, 7.52)$
 $= 7.5200 \text{ mm}$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 52.2817 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 5358.7075 mm

Input, Nozzle Desc: K675

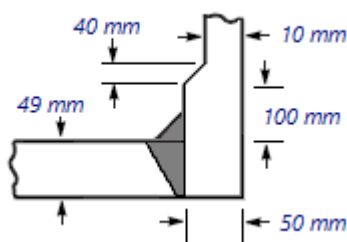
From: 60

Pressure for Reinforcement Calculations	P	4621.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	454	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	137.29	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	5398.00	mm
Head Finished (Minimum) Thickness	t	49.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	126.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		76.2000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	2000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	49.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K675

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	76.200	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0)$
= 45.6284 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4621.0 \cdot 41.1 / (126 \cdot 1.0 - 0.6 \cdot 4621.0)$
= 1.5413 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.9358 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 40.0/0.5773
= 69.2821 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

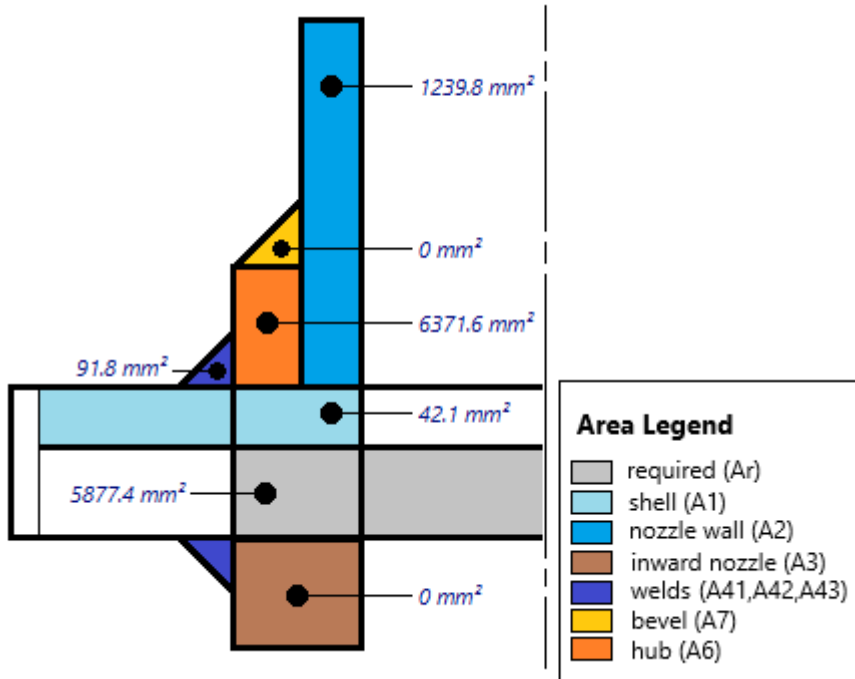
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	242.1554	mm
Parallel to Vessel Wall, opening length	d	121.0777	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 126.0/137.3)
= 0.918

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 126.0/137.3)
= 0.918

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.918, 1.0)
= 0.918



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	58.774	5.870	NA
Area in Shell	A1	0.421	41.808	NA
Area in Nozzle Wall	A2	12.398	13.774	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.918	0.918	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	63.716	63.716	NA
TOTAL AREA AVAILABLE	Atot	77.453	120.215	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 42.76 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (121.0777 \cdot 45.6284 \cdot 1.0 + 2 \cdot 47.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 58.774 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 121.078(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 47.0$
 $(1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918)$
 $= 0.421 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 86.78)(7.0 - 1.54)0.9178/\sin(44.5)$
 $= 12.398 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9178 + (0.0)^2 \cdot 0.9178$



= 0.918 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(86.8, 2000.0, 100.0)) \cdot (50.0 - 10.0) \cdot 0.918 \\ &= 63.716 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 10.0, tr = 1.405, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.201, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -16 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4621.00/10340.00 = 0.447

Weld Size Calculations, Description: K675

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (58.7735 - 0.4212 + 2 • 47.0 • 0.9178 •
(1.0 • 46.0 - 45.6284))137)
= 82139.95 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (76.1136 + 0.9178 - 0.0 • 0.92) • 137
= 107841.06 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (76.1136 + 0.0 + 0.9178 + (39.6836)) • 137
= 163396.62 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (76.1136 + 0.0 + 0.9178 + 0.0 + (39.6836)) • 137
= 163396.62 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 259.5364 \cdot 10.0 \cdot 0.49 \cdot 126 \\ &= 25666. \text{ Kg} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 95.1535) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 126361. \text{ Kg} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 259.5364 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 194278. \text{ Kg} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (25666 + 126361) = 152027 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (25666 + 0 + 194278 + 0) = 219944 \text{ Kg} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (25666 + 194278 + 0) = 219944 \text{ Kg} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 152026 Kg, must exceed W = 82139 Kg	or W1 = 107841 Kg
Path 2-2 = 219944 Kg, must exceed W = 82139 Kg	or W2 = 163396 Kg
Path 3-3 = 219944 Kg, must exceed W = 82139 Kg	or W3 = 163396 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K675

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	76.200 mm.
Actual Thickness Used in Calculation	10.000 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 4621.0 \cdot 2702.0 / (2 \cdot 137 \cdot 1.0 - 0.2 \cdot 4621.0) \\ &= 45.6284 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 4621.0 \cdot 41.1 / (126 \cdot 1.0 - 0.6 \cdot 4621.0) \\ &= 1.5413 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.9358 mm

Intermediate Hub Nozzle Calculations:

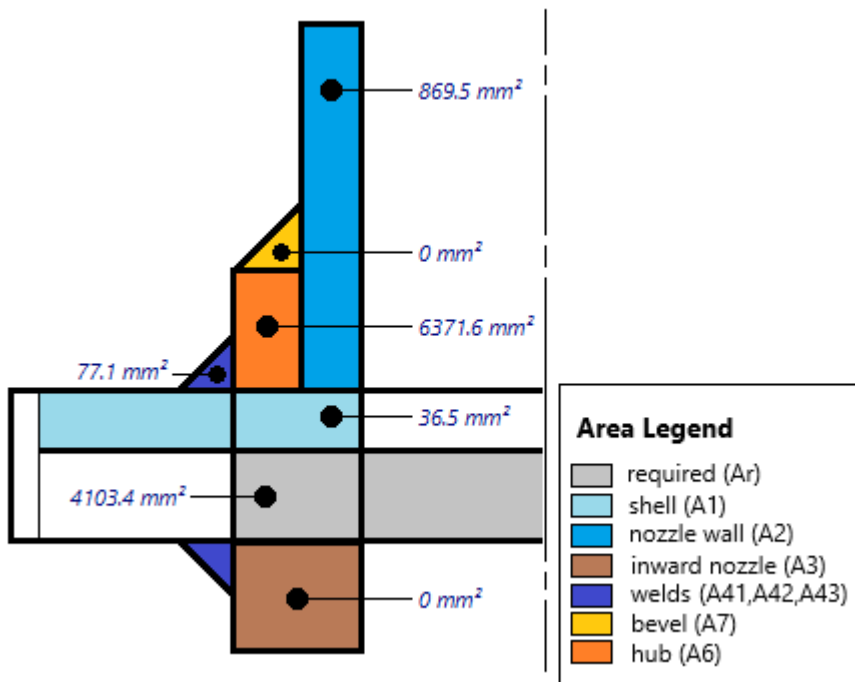
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0/0.5773$
 $= 69.2821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	188.2000	mm
Parallel to Vessel Wall	Rn+tn+t	94.1000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	86.7821	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	41.034	4.098	NA
Area in Shell	A1	0.365	36.247	NA
Area in Nozzle Wall	A2	8.695	9.660	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.771	0.771	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	63.716	63.716	NA
TOTAL AREA AVAILABLE	Atot	73.547	110.393	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (82.2 \cdot 45.6284 \cdot 1.0 + 2 \cdot 47.0 \cdot 45.6284 \cdot 1.0 \cdot (1 - 0.92))$
 $= 41.034 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 106.0(1.0 \cdot 46.0 - 1.0 \cdot 45.628) - 2 \cdot 47.0 \\ &\quad (1.0 \cdot 46.0 - 1.0 \cdot 45.6284) \cdot (1 - 0.918) \\ &= 0.365 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot tlnp)(tn - trn)fr2 \\ &= (2 \cdot 86.78)(7.0 - 1.54) \cdot 0.9178 \\ &= 8.695 \text{ cm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi\text{-}can/0.707)^2 - \text{Area Lost} \cdot fr2 \\ &= (10.0^2 - 0.16) \cdot 0.9178 + (0.0^2 - 0.0) \cdot 0.9178 \\ &= 0.771 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(86.8, 2000.0, 100.0)) \cdot (50.0 - 10.0) \cdot 0.918 \\ &= 63.716 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5413 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 48.6284 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 48.6284 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0178 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.02, \max(48.6284, 4.5)] \\ &= 8.0200 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.5413, 8.02) \\ &= 8.0200 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: K675

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv) \\ &= \max(0, (41.0343 - 0.3651 + 2 \cdot 47.0 \cdot 0.9178 \cdot \\ &\quad (1.0 \cdot 46.0 - 45.6284))137) \\ &= 57384.02 \text{ Kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi\text{-}Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (72.4106 + 0.7709 - 0.0 \cdot 0.92) \cdot 137 \\ &= 102451.35 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (72.4106 + 0.0 + 0.7709 + (39.6836)) \cdot 137 \end{aligned}$$



= 158006.94 Kgf

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\ &= (72.4106 + 0.0 + 0.7709 + 0.0 + (39.6836)) \cdot 137 \\ &= 158006.94 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 176.2 \cdot 10.0 \cdot 0.49 \cdot 126 \\ &= 17425. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 64.6) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 126 \\ &= 85787. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 176.2 \cdot (49.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 131896. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (17425 + 85787) = 103211 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (17425 + 0 + 131896 + 0) = 149321 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (17425 + 131896 + 0) = 149321 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 103211 Kgf, must exceed W = 57384 Kgf	or W1 = 102451 Kgf
Path 2-2 = 149320 Kgf, must exceed W = 57384 Kgf	or W2 = 158006 Kgf
Path 3-3 = 149320 Kgf, must exceed W = 57384 Kgf	or W3 = 158006 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4659 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 102.2476 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 2174.4441 mm

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